1 Counting principles

In this chapter you will learn:

- how to break down complicated questions into parts that are easier to count, and then combine them together
- how to count the number of ways to arrange a set of objects
- the algebraic properties of a useful new tool called the factorial function
- in how many ways you can choose objects from a group
- strategies for applying these tools to harder problems.

Introductory problem

If a computer can print a line containing all 26 letters of the alphabet in 0.01 seconds, estimate how long it would take to print all possible permutations of the alphabet.

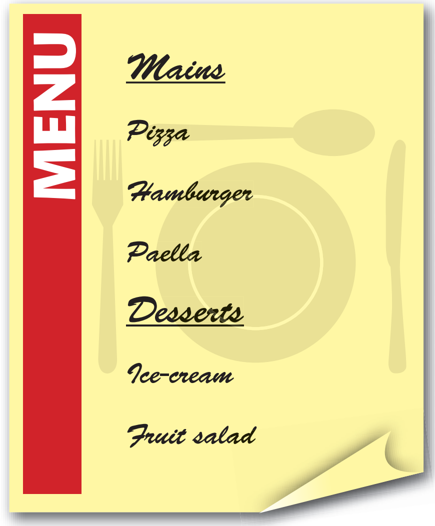
Counting is one of the first things we learn in mathematics and at first it seems very simple. If you were asked to count how many people there are in your school, this would not be too tricky. If you were asked how many chess matches would need to be played if everyone were to play everyone else, this would be a little more complicated. If you were asked how many different football teams could be chosen, you might find that the numbers become far too large to count without coming up with some clever tricks. This chapter aims to help develop strategies for counting in such difficult situations.

Counting sometimes gets extremely difficult. Are there more whole numbers or odd numbers; fractions or decimals? Have a look at the work of Georg Cantor, and the result may surprise you!

1A The product principle and the addition principle

Counting very small groups is easy. So, we need to break down more complicated problems into counting small groups. But how do we then combine these together to come up with an answer to the overall problem? The answer lies in using the product principle and the addition principle, which can be illustrated using the following menu.

The word analysis literally means 'breaking up'. When a problem is analysed it is broken down into simpler parts. One of the purposes of studying mathematics is to develop an analytical mind, which is considered very useful in many different disciplines.



Anna would like to order a main course *and* a dessert. She can choose one of three main courses and one of two desserts. How many different choices could she make? Bob would like to order *either* a main course *or* a dessert. He can choose one of the three main courses or one of the two desserts; how many different orders can he make?

$n(A)$ means the size of the set of options of A. See Prior Learning Section G on the CD-ROM.

We can use the notation $n(A)$ to represent the number of ways of making a choice about A.

The **product principle** tells us that when we want to select one option from A *and* one option from B we *multiply* the individual possibilities together.

KEY POINT 1.1

The Product principle (AND rule)

The number of ways of in which both choice A *and* choice B can be made is the product of the number of options for A and the number of options for B.

$n(A \text{ AND } B) = n(A) \times n(B)$

The **addition principle** tells us that when we wish to select one option from A *or* one option from B we *add* the individual possibilities together.

The addition principle has one essential restriction. You can only use it if there is no overlap between the choices for A and the choices for B. For example, you cannot apply the addition principle to counting the number of ways of getting an odd number or a prime number on a die. If there is no overlap between the choices for A and for B, the two events are **mutually exclusive.**

KEY POINT 1.2

The Addition Principle (OR rule)

The number of ways of in which either choice A *or* choice B can be made is the sum of the number of options for A and the number of options for B.

If A and B are mutually exclusive then
$n(A \text{ OR } B) = n(A) + n(B)$

The hardest part of applying either the addition or product principle is breaking the problem down and deciding which principle to use. You must make sure that you have included all of the cases and checked that they are mutually exclusive. It is often useful to rewrite questions to emphasise what is required, 'AND' or 'OR'.

Worked example 1.1

An examination has ten questions in section A and four questions in section B. How many different ways are there to choose questions if you must:

(a) choose one question from each section?
(b) choose a question from either section A or section B?

Describe the problem accurately

(a) Choose one question from A (10 ways)
AND
one from B (4 ways)

'AND' means we should apply the product principle: $n(A) \times n(B)$

Number of ways $= 10 \times 4$
$= 40$

Describe the problem accurately

(b) Choose one question from A (10 ways)
OR
one from B (4 ways)

'OR' means we should apply the addition principle: $n(A) + n(B)$

Number of ways $= 10 + 4$
$= 14$

In the example above we cannot answer a question twice so there are no repeated objects; however, this is not always the case.

Worked example 1.2

In a class there are awards for best mathematician, best sportsman and nicest person. Students can receive more than one award. In how many ways can the awards be distributed if there are twelve people in the class?

Describe the problem accurately

Choose one of 12 people for the best mathematician (12 ways)
AND
one of the 12 for best sportsmen (12 ways)
AND
one of the 12 for nicest person. (12 ways)

Apply the product principle

$12 \times 12 \times 12 = 1728$

This leads us to a general idea.

KEY POINT 1.3

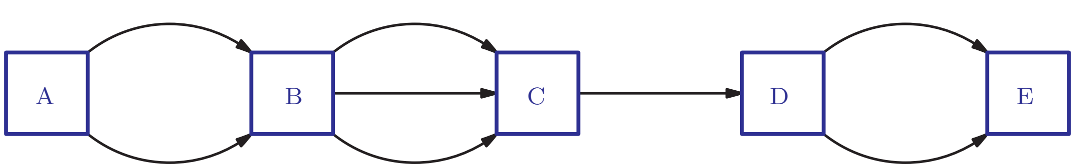
The number of ways of selecting something r times from n objects is n^r.

Exercise 1A

1. If there are 10 ways of doing A, 3 ways of doing B and 19 ways of doing C, how many ways are there of doing
   (a) (i) both A and B? (ii) both B and C?
   (b) (i) either A or B? (ii) either A or C?

2. If there are 4 ways of doing A, 7 ways of doing B and 5 ways of doing C, how many ways are there of doing
   (a) all of A, B and C?
   (b) exactly one of A, B or C?

3. How many different paths are there
   (a) from A to C?
   (b) from C to E?
   (c) from A to E?



4. Jamil is planting out his garden and needs one new rose bush and some dahlias. There are 12 types of rose and 4 varieties of dahlia in his local nursery. How many possible selections does he have to choose from? *[3 marks]*

5. A lunchtime menu at a restaurant offers 5 starters, 6 main courses and 3 desserts. How many different choices of meal can you make if you would like

(a) a starter, a main course and a dessert?

(b) a main course and either a starter or a dessert?

(c) any two different courses? *[6 marks]*

6. Five men and three women would like to represent their club in a tennis tournament. In how many ways can one mixed doubles pair be chosen? *[3 marks]*

7. A mathematics team consists of one student from each of years 7, 8, 9 and 10. There are 58 students in year 7, 68 in year 8, 61 in year 9 and 65 in year 10.

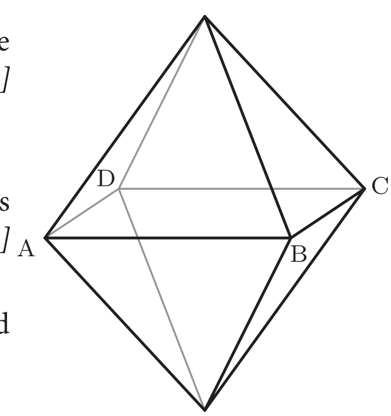
(a) How many ways are there of picking the team?

Year 10 is split into three classes: 10A (21 students), 10B (23 students) and 10C (21 students).

(b) If students from 10B cannot participate in the competition, how many ways are there of picking the team? *[4 marks]*

8. Student passwords consist of three letters chosen from A to Z, followed by four digits chosen from 1–9. Repeated characters are allowed. How many possible passwords are there? *[4 marks]*

9. A beetle walks along the edges from the base to the tip of an octahedral sculpture, visiting exactly two of the middle vertices (A, B, C or D). How many possible routes are there? *[6 marks]*



10. Professor Square has 15 different ties (seven blue, three red and five green), four waistcoats (red, black, blue and brown) and 12 different shirts (three each of red, pink, white and blue). He always wears a shirt, a tie and a waistcoat.

(a) How many different outfits can he make?

Professor Square never wears any outfit that combines red with pink.

(b) How many different outfits can he make with this limitation? *[6 marks]*

11. How many different three digit numbers can be formed using the digits 1,2,3,5,7

(a) once only?

(b) if digits can be repeated? *[6 marks]*

12. In how many ways can

(a) four toys be put into three boxes?

(b) three toys be put into five boxes? *[6 marks]*

If you look in other textbooks you may see permutations referred to in other ways. This is an example of a word that can take slightly different meanings in different countries. The definition here is the one used in the International Baccalaureate® (IB).

$n!$ occurs in many other mathematical situations. You will see it in the Poisson distribution (see Section 23D), and if you study option 9 on calculus, you will see that it is also important in a method for approximating functions called Taylor series.

1B Counting arrangements

The word 'ARTS' and the word 'STAR' both contain the same letters, but arranged in a different order. They are both arrangements (also known as **permutations**) of the letters R, A, T and S. We can count the number of different permutations.

There are four possibilities for the first letter, then for each choice of the first letter there are three options for the second letter (because one of the letters has already been used). This leaves two options for the third letter and then the final letter is fixed. Using the 'AND rule' the number of possible permutations is $4 \times 3 \times 2 \times 1 = 24$.

The number of permutations of n different objects is equal to the product of all positive integers less than or equal to n. This **expression** is abbreviated to $n!$ (pronounced 'n factorial').

KEY POINT 1.4

The number of ways of arranging n objects is $n!$

$n! = n(n-1)(n-2)\ldots \times 2 \times 1$

Worked example 1.3

A test has 12 questions. How many different arrangements of the questions are possible?

Describe the problem accurately

Permute (arrange) 12 items
Number of permutations $= 12!$
$= 479\,001\,600$

In examination questions you might have to combine the idea of permutations with the product and addition principles.

Worked example 1.4

A seven-digit number is formed by using each of the digits 1–7 exactly once. How many such numbers are even?

Describe the problem accurately: even numbers end in 2, 4 or 6	Pick the final digit to be even (3 ways)
Only 6 digits left to arrange	AND then permute the remaining 6 digits (6! ways)
Apply the product principle	$3 \times 720 = 2160$ possible even numbers

This example shows a very common situation where there is a constraint, in this case we have to end with an even digit. It can be more efficient to fix each part of the constraint separately, instead of searching all the possibilities for the ones which are allowed.

EXAM HINT

Factorials get very large very quickly. Although you should know factorials up to 6!, most of the time you will use your calculator. See Calculator skills sheet 3 on the CD-ROM.

Worked example 1.5

How many permutations of the word SQUARE start with three vowels?

Describe the problem accurately	Permute the three vowels at the beginning (3! ways) AND Permute the three consonants at the end (3! ways)
Apply the product principle	Number of ways $= 3! \times 3! = 36$

Exercise 1B

1. Evaluate:

   (a) (i) $5!$ (ii) $6!$

   (b) (i) $2 \times 4!$ (ii) $3 \times 5!$

   (c) (i) $6! - 5!$ (ii) $6! - 4 \times 5!$

2. Evaluate:

   (a) (i) $8!$ (ii) $11!$

   (b) (i) $9 \times 5!$ (ii) $9! \times 5$

   (c) (i) $12! - 10!$ (ii) $9! - 7!$

3. Find the number of ways of arranging:

   (a) 6 objects (b) 8 objects (c) 26 objects.

4. (a) How many ways are there of arranging seven textbooks on a shelf?

   (b) In how many of those arrangements is the single biggest textbook not at either end? *[5 marks]*

5. (a) How many five-digit numbers can be formed by using each of the digits 1–5 exactly once?

   (b) How many of those numbers are divisible by 5? *[5 marks]*

6. A class of 16 pupils and their teacher are queuing outside a cinema.

   (a) How many different arrangements are there?

   (b) How many different arrangements are there if the teacher has to stand at the front? *[5 marks]*

7. A group of nine pupils (five boys and four girls) are lining up for a photograph, with all the girls in the front row and all the boys at the back. How many different arrangements are there? *[5 marks]*

8. (a) How many six-digit numbers can be made by using each of the digits 1–6 exactly once?

   (b) How many of those numbers are smaller than 300 000? *[5 marks]*

9. A class of 30 pupils are lining up in three rows of ten for a class photograph.
How many different arrangements are possible? *[6 marks]*

10. A baby has nine different toy animals. Five of them are red and four of them are blue. She arranges them in a line so that the colours are arranged symmetrically. How many different arrangements are possible? *[7 marks]*

1C Algebra of factorials

To solve more complicated counting problems we often need to simplify expressions involving factorials. This is done using the formula for factorials, which you saw in Key point 1.4:

$$n! = n(n-1)(n-2)\ldots\times 2\times 1$$

Worked example 1.6

(a) Evaluate $9! \div 6!$

(b) Simplify $\dfrac{n!}{(n-3)!}$

(c) Write $10 \times 11 \times 12$ as a ratio of two factorials.

Write in full and look for common factors in denominator and numerator

(a) $\dfrac{9\times 8\times 7\times 6\times 5\ldots\times 2\times 1}{6\times 5\ldots\times 2\times 1}$

$= 9\times 8\times 7$

$= 504$

Write in full and look for common factors in denominator and numerator

(b) $\dfrac{n\times(n-1)\times(n-2)\times(n-3)\ldots\times 2\times 1}{(n-3)\times(n-4)\ldots\times 2\times 1}$

$= n(n-1)(n-2)$

Reverse the ideas from (b)

(c) $10\times 11\times 12 = \dfrac{1\times 2\ldots\times 9\times 10\times 11\times 12}{1\times 2\ldots\times 9}$

$= \dfrac{12!}{9!}$

You can usually solve questions involving sums or differences of factorials by looking for common factors of the terms. It is important to understand the link between one factorial and the next:

KEY POINT 1.5

$$(n+1)! = (n+1) \times n!$$

Worked example 1.7

Simplify:

(a) $9! - 7!$ (b) $(n+1)! - n! + (n-1)! = 72 \times 7!$

Link 9! and 7!

(a) $9! = 9 \times 8! = 9 \times 8 \times 7!$

Take out the common factor of 7!

$$9! - 7! = 72 \times 7! - 7 \\ = (72-1) \times 7! = 71 \times 7!$$

Link $(n+1)!$, $n!$ and $(n-1)!$

(b) $n! = n \times (n-1)!$

$$(n+1)! = (n+1) \times n! = (n+1) \times n \times (n-1)!$$

Take out the common factor of $(n-1)!$

$$(n+1)! - n! + (n-1)! = n(n+1)(n-1)! - n(n-1)! + (n-1)!$$
$$= (n-1)!(n^2 + n - n + 1) = (n-1)!(n^2+1)$$

0! is defined to be 1. Why might this be?

$\frac{1}{2}!$ is $\frac{\sqrt{\pi}}{2}$

To explore why you need to look at the Gamma function.

Being able to simplify expressions involving $n!$ is useful because $n!$ becomes very large very quickly. Often factorials cannot be evaluated even using a calculator. For example, a standard scientific calculator can only calculate up to $69! = 1.71 \times 10^{98}$ (to 3 significant figures).

EXAM HINT

Remember that you will lose one mark per paper in the IB if you give any answers to less than three significant figures (3SF) (unless the answer is exact of course!). See Prior learning Section B on the CD-ROM if you need a reminder about significant figures.

Exercise 1C

1. Fully simplify the following fractions:

(a) (i) $\frac{7!}{6}$ (ii) $\frac{12!}{11!}$

(b) (i) $\frac{8!}{6!}$ (ii) $\frac{11!}{8!}$

(c) (i) $\dfrac{100!}{98!}$ (ii) $\dfrac{51!}{49!}$

(d) (i) $\dfrac{n!}{(n-1)!}$ (ii) $\dfrac{(a+1)!}{a!}$

(e) (i) $\dfrac{a!}{(a-2)!}$ (ii) $\dfrac{(b+1)!}{(b-1)!}$

(f) (i) $\dfrac{(x+6)!}{(x+8)!}$ (ii) $\dfrac{(x-7)!}{(x-4)!}$

2. Write these as a ratio of 2 factorials:

(a) (i) $7 \times 8 \times 9$ (ii) $6 \times 5 \times 4 \times 3$

(b) (i) $1013 \times 1014 \times 1015 \times 1016$

(ii) $307 \times 308 \times 309$

(c) (i) $(n+2)(n+3)(n+4)(n+5)$

(ii) $n(n-1)(n+1)(n+2)$

3. Explain why $6! - 5! = 5 \times 5!$

Simplify in a similar fashion:

(a) (i) $10! - 9!$ (ii) $12! - 10!$

(b) (i) $14! - 3 \times 13!$ (ii) $16! + 5 \times 15!$

(c) (i) $11! + 11 \times 9!$ (ii) $12! - 22 \times 10!$

(d) (i) $n! - (n-1)!$ (ii) $(n+2)! - n!$

4. Solve $\dfrac{n!}{(n-2)!} = 20$ where n is a positive integer. *[5 marks]*

5. Solve $\dfrac{(n+1)!}{(n-2)!} = 990$. *[6 marks]*

6. Solve $n! - (n-1)! = 16(n-2)!$ for $n \in \mathbb{N}$. *[6 marks]*

1D Counting selections

Suppose that three pupils are to be selected from a class of 11 to attend a meeting with the Head Teacher. How many different groups of three can be chosen?

In this example we need to *choose* three pupils out of 11, but they are not to be arranged in any specified order. The order is not important; the selection of Ali, Bill and then Cathy is the same as the selection of Bill, Cathy and then Ali. This sort of selection is called a **combination**. In general, the formula for

the number of ways of choosing r objects out of n is given the symbol $\binom{n}{r}$, or ${}^nC_r$ (pronounced 'n C r' or 'n choose r').

EXAM HINT

The Formula booklet tells you the formula, but not what it is used for.

KEY POINT 1.6

The number of ways of choosing r objects out of n is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Does learning the terminology of $n!$ and $\binom{n}{r}$ give you new mathematical knowledge?

Does this mathematical language clarify or muddle the principles behind it? Is it necessary to have 'expert language' in all subjects? Does giving something a label help you think about it?

Worked example 1.8

A group of 12 friends want to form a team for a five-a-side football tournament.

(a) In how many different ways can a team of five be chosen?

(b) Rob and Amir are the only goalkeepers and they cannot play in any other position. If the team of five has to contain exactly one goalkeeper, how many different ways can the team be chosen?

Describe the problem accurately

(a) Choosing 5 players from 12

Number of ways $= \binom{12}{5} = 792$

Describe the problem accurately

(b) Pick a goalkeeper, then fill in the rest of the team

Split into cases to satisfy the condition

Amir is in goal (1 way)

AND

Choose four other players (? ways)

OR

Rob is in goal. (1 way)

AND

Choose four other players (? ways)

With Amir picked there are four remaining slots, but Rob cannot be picked so ten players remain

Number of ways with Amir in goal $= \binom{10}{4}$

Number of ways with Rob in goal $= \binom{10}{4}$

Apply the addition principle and product principle

Total number of ways $= 1 \times \binom{10}{4} + 1 \times \binom{10}{4}$

$= 210 + 210$

$= 420$

EXAM HINT

You can calculate $\binom{n}{r}$ on your calculator using a button like this; see Calculator skills sheet 3 on the CD-ROM.

We will use the idea of combinations to help us expand brackets in chapter 8 and to find probabilities in Section 23C.

Exercise 1D

1. Evaluate:

(a) (i) $\binom{7}{2}$ (ii) $\binom{12}{5}$

(b) (i) $3\times\binom{6}{3}$ (ii) $10\times\binom{6}{5}$

(c) (i) $\binom{5}{0}\times\binom{9}{5}$ (ii) $\binom{10}{8}\times\binom{3}{1}$

(d) (i) $\binom{5}{2}+\binom{9}{5}$ (ii) $\binom{6}{0}+\binom{7}{3}$

2. (a) (i) In how many ways can six objects be selected from eight?

(ii) In how many ways can five objects be selected from nine?

(b) (i) In how many ways can either three objects be selected from ten, or seven objects be selected from 12?

(ii) In how many ways can either two objects be selected from five, or three objects be selected from four?

(c) (i) In how many ways can five objects be selected from seven and three different objects be selected from eight?

(ii) In how many ways can six objects be selected from eight and three different objects be selected from seven?

3. Solve:

(a) (i) $\binom{n}{2}=91$ (ii) $\binom{n}{2}=351$

(b) (i) $\binom{n}{3}=1330$ (ii) $\binom{n}{3}=680$

4. An exam paper consists of 15 questions. Students can select any nine questions to answer. How many different selections can be made? *[4 marks]*

5. Suppose you are revising for seven subjects, but you can only complete three in one evening.

(a) In how many ways can you select three subjects to do on Monday evening?

(b) If you have to revise mathematics on Tuesday, in how many ways can you select the subjects to do on Tuesday evening? *[5 marks]*

6. In the 'Pick'n'Mix' lottery, players select seven numbers out of 39. How many different selections are possible? *[4 marks]*

7. There are 16 boys and 12 girls in a class. Three boys and two girls are needed to take part in the school play. In how many different ways can they be selected? *[5 marks]*

8. A soccer team consists of one goalkeeper, four defenders, four midfielders and two forwards. A manager has three goalkeepers, eight defenders, six midfielders and five forwards in the squad. In how many ways can she pick the team? *[5 marks]*

9. A school is planning some trips over the summer. There are 12 places on the Greece trip, ten places on the China trip and ten places on the Disneyland trip. There are 140 pupils in the school who are all happy to go on any of the three trips.

In how many ways can the spaces be allocated? *[5 marks]*

10. A committee of three boys and three girls is to be selected from a class of 14 boys and 17 girls. In how many ways can the committee be selected if:

(a) Ana has to be on the committee?

(b) the girls must include either Roberta or Priya, but not both? *[6 marks]*

11. Raspal's sweet shop stocks seven different types of 2¢ sweets and five different types of 5¢ sweets. If you want no more than one of each sweet, how many different selections of sweets can be made when spending:

(a) exactly 6¢?

(b) exactly 7¢?

(c) exactly 10¢?

(d) at most 5¢? *[6 marks]*

12. An English examination has two sections. Section A has five questions and section B has four questions. Four questions must be answered in total.

(a) How many different ways are there of selecting four questions to answer if there are no restrictions?

(b) How many different ways are there of selecting four questions if there must be at least one question answered from each section? *[6 marks]*

13. 15 points lie around a circle. Each point is connected to every other point by a straight line. How many lines are formed in this way? *[4 marks]*

14. Ten points are drawn on a sheet of paper so that no three lie in a straight line. By connecting up the points:

(a) how many different triangles can be drawn?

(b) how many different quadrilaterals can be drawn? *[7 marks]*

15. At a party, everyone shakes hands with everyone else. If 276 handshakes are exchanged, how many people are there at the party? *[6 marks]*

16. A group of 45 students are to be seated in three rows of 15 for a school photograph. Within each row, students must sit in alphabetical order according to name, but there is no restriction determining the row in which a student must sit.

How many different seating arrangements are possible, assuming no students have identical names? *[6 marks]*

1E Exclusion principle

The **exclusion principle** is a way to count what you are interested in by first counting what you are *not* interested in. This is often needed for counting where a certain property is prohibited (not allowed).

KEY POINT 1.7

The Exclusion principle

Count what you are not interested in and subtract it from the total.

Imagine that a five-digit code is formed using each of the digits 1–5 exactly once. If we wanted to count how many such codes do *not* end in '25' we could work out *all* of the possible options that do not end in '25':

Worked example 1.9a

How many five-digit codes formed by using each of the digits 1–5 exactly once do not end in '25'?

Describe the problem accurately

Pick the final digit from {1, 2, 3, 4} (4 ways)
AND
arrange the remaining four digits (4! ways)
OR
Pick the final digit as 5 (1 way)
AND
pick the penultimate digit from {1, 3, 4} (3 ways)
AND
arrange the remaining three digits (3! ways)

Use the product and addition principles

$(4 \times 4!) + (1 \times 3 \times 3!) = 114$

This method works, but it is long and complicated. An easier way is to use the exclusion principle.

Worked example 1.9b

How many five-digit codes formed by using each of the digits 1–5 exactly once do not end in '25'?

Describe the problem accurately

Count permutations of five digit codes (5! ways)
then EXCLUDE cases where
the last two digits are '25' (1 way)
AND
arrange the remaining three digits (3! ways)

Use the product and exclusion principles

$5! - 1 \times 3! = 114$

Many questions ask us to use the exclusion principle in a situation containing an 'at least' or 'at most' restriction.

Worked example 1.10

Theo has eight different jigsaws and five different toy bears. He chooses four things to play with. In how many ways can he make a selection with at least two bears?

Describe the problem accurately

Count combinations of four toys from thirteen $\binom{13}{4}$ ways

continued . . .

Then EXCLUDE

Combinations with no bears $\binom{5}{0}$ ways

AND

Four jigsaws $\binom{8}{4}$ ways

OR

Combinations with one bear $\binom{5}{1}$ ways

AND

Three jigsaws $\binom{8}{3}$ ways

Apply the product and addition principles

Number of ways $= \binom{13}{4} - \left(\binom{5}{0}\binom{8}{4} + \binom{5}{1}\binom{8}{3}\right)$

$= 714 - (1 \times 70 + 5 \times 56) = 365$

EXAM HINT

This example highlights the ambiguity of the English language. Can you see why we exclude (no bears and four jigsaws) OR (one bear and three jigsaws) rather than (no bears and four jigsaws) AND (one bear and three jigsaws)?

Be careful when you read questions. For example, the opposite of a word beginning in a consonant OR ending in a consonant is not a word beginning in a vowel OR ending in a vowel. For example, the word 'LOVE' fits both descriptions. If you are interested there is formal work on this, called De Morgan's laws, in the Sets, Relations and Groups option (option 8).

Exercise 1E

1. How many of the numbers between 101 and 800 inclusive are not divisible by 5? *[4 marks]*

2. How many permutations of the letters of the word JUMPER do not start with a J? *[5 marks]*

3. A bag contains 12 different chocolates, four different mints and six different toffees. Three sweets are chosen. How many ways are there of choosing

(a) all not chocolates? (b) not all chocolates? *[6 marks]*

4. How many permutations of the letters of KITCHEN

(a) do not begin with KI?

(b) do not have K and I in the first two letters? *[6 marks]*

5. A committee of five people is to be selected from a class of 12 boys and nine girls. How many such committees include at least one girl? *[6 marks]*

6. In a word game there are 26 letter tiles, each with a different letter. How many ways are there of choosing seven tiles so that at least two are vowels? *[6 marks]*

7. A committee of six is to be selected from a group of ten men and 12 women. In how many ways can the committee be chosen if it has to contain at least two men and one woman? *[6 marks]*

8. Seven numbers are chosen from the integers 1–19 inclusive. How many have

(a) at most two even numbers?

(b) at least two even numbers? *[7 marks]*

9. How many permutations of the letters DANIEL do not begin with D or do not end with L? *[6 marks]*

1F Counting ordered selections

Sometimes we want to choose a number of objects from a bigger group but the order in which they are chosen *is important*. For example, finding the possibilities for the first three finishers in a race or forming numbers from a fixed group of digits. The strategy for dealing with these situations is first to choose from the larger group and then permute the chosen objects.

Worked example 1.11

A class of 28 pupils has to select a committee consisting of a class representative, a treasurer, a secretary and a football captain. Each post needs to be a different person. In how many different ways can the four posts be filled?

Select four people and then allocate them to different jobs. Note that this is the same as selecting four people and then permuting them	Choose 4 from 28 $\binom{28}{4}$ ways AND Permute those 4 (4! ways)
Apply the product principle	Number of ways $=\binom{28}{4}\times 4! = 491400$

Suppose that the whole set has size n and we are selecting and ordering a subset of size r. The number of ways of doing this is given the symbol ${}^nP_r$ and is described as 'the number of permutations of r objects out of n'. We can use the formula for $\binom{n}{r}$ to deduce a formula for ${}^nP_r$: there are $\binom{n}{r}$ ways of choosing the objects in the subset, and these can be arranged in $r!$ ways. Therefore, ${}^nP_r = \binom{n}{r} \times r!$ or:

KEY POINT 1.8

$${}^nP_r = \frac{n!}{(n-r)!}$$

This can be written, using factorial algebra, to:

$${}^nP_r = n(n-1)(n-2)\ldots(n-r+1)$$

Written in this form you can see that the formula for ${}^nP_r$ is an application of the 'AND rule'. For example, suppose that there are five people in a race and we want to count the number of possibilities for the first three positions. We can reason that if there are five options for the winner then, for each winner, there are four options for the second place and three options for the third place. This gives $5 \times 4 \times 3 = {}^5P_3$.

Exercise 1F

1. Evaluate:

   (a) (i) ${}^6P_1$ (ii) ${}^5P_1$

   (b) (i) ${}^8P_2$ (ii) ${}^{11}P_2$

   (c) (i) ${}^{10}P_3$ (ii) ${}^{12}P_3$

2. Find the number of permutations of:

   (i) 4 objects out of 10 (ii) 6 objects out of 7.

3. Find the number of ways of selecting when order matters:

   (i) 3 objects out of 5 (ii) 2 objects out of 15.

4. Solve the following equations:

(a) (i) ${}^{n}P_2 = 42$ (ii) ${}^{n}P_2 = 90$

(b) (i) ${}^{n}P_3 = 9 \times {}^{n}P_2$ (ii) ${}^{n}P_3 = 12 \times {}^{n}P_2$

5. In a 'Magic Sequence' lottery draw there are 39 balls numbered 1–39. Seven balls are drawn at random. The result is a sequence of seven numbers which must be matched in the correct order to win the grand prize. How many possible sequences can be made? *[4 marks]*

6. A teacher needs to select four pupils from a class of 24 to receive four different prizes. How many possible ways are there to award the prizes? *[4 marks]*

7. How many three-digit numbers can be formed from digits 1–9 if no digit can be repeated? *[4 marks]*

8. Eight athletes are running a race. In how many different ways can the first three places be filled? *[4 marks]*

9. An identification number consists of two letters followed by four digits chosen from 0–9. No digit or letter may appear more than once. How many different identification numbers can be made? *[6 marks]*

10. Prove that ${}^{n}P_{n-1} = {}^{n}P_{n}$. *[3 marks]*

11. Three letters are chosen from the word PICTURE and arranged in order. How many of the possible permutations contain at least one vowel? *[6 marks]*

12. Eight runners compete in a race. In how many different ways can the three medals be awarded if James wins either a gold or a silver? *[6 marks]*

13. A class of 18 needs to select a committee consisting of a President, a Secretary and a Treasurer. How many different ways are there to form the committee if Rajid does not want to be President? *[6 marks]*

14. Solve the equation ${}^{2n}P_3 = {}^{6}P_n$, justifying that you have found all solutions. *[6 marks]*

1G Keeping objects together or separated

How many letter arrangements of the word 'SQUARE' have the Q and U next to each other? How many have the vowels all separated?

When a problem has constraints like this we need some clever tricks to deal with them.

The first type of problem we will look at is where objects are forced to stay together. The trick is to imagine the letters in the word SQUARE as being on tiles. If the Q and the U need to be together, we are really dealing with five tiles, one containing QU:

We must also remember that the condition is satisfied if Q and U in are in the other order.

Is mathematics about finding answers or having strategies and ideas? Has the balance between the two changed as you have progressed through learning mathematics?

Worked example 1.12

How many permutations of the word 'SQUARE' have the Q and the U next to each other?

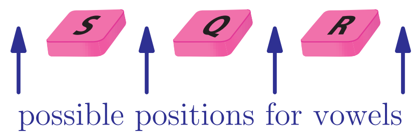
Describe the problem accurately	Find the number of permutations of the five 'tiles' (S, QU, A, R, E) (5! ways) AND Find the number of permutations of the letters QU on the 'double tile' (2! ways)
Apply the product principle	Number of ways $= 5! \times 2! = 240$

KEY POINT 1.9

If a group of items have to be kept together, treat the items as one object. Remember that there may be permutations of the items within this group too.

Another sort of constraint is where objects have to be kept apart. This is *not* the opposite of objects being kept together unless we are separating only two objects. For example, the opposite of three objects staying together includes having two of them together and the third one apart. So when dealing with keeping objects apart we need to focus on the gaps that the critical objects can fit into.

Consider the question of how many permutations of the word SQUARE have none of the vowels together. We first permute all of the consonants. One such permutation is



There are four gaps in which we can put the vowels. We only have three vowels and so only need to choose three of the gaps, and then decide in what order to insert the vowels.

Worked example 1.13

How many permutations of the word 'SQUARE' have none of the vowels together?

Describe the problem accurately

Permute 3 consonants — $3!$ ways

AND

Choose 3 out of the four gaps — $\binom{4}{3}$ ways

AND

Permute the three vowels to put into the gaps — $3!$ ways

Apply the product principle

Total number of ways $3! \times 4 \times 3! = 144$

KEY POINT 1.10

If k objects have to be kept apart, permute all the other objects and then count the permutations of k 'gaps'.

Exercise 1G

1. In how many ways can 14 people be arranged in a line if Joshua and Jolene have to stand together? *[6 marks]*

2. Students from three different classes are standing in the lunch queue. There are six students from 10A, four from 10B and four from 10C. In how many ways can the queue be arranged if students from the same class have to stand together? *[6 marks]*

3. In how many ways can six Biology books and three Physics books be arranged on the shelf if the three Physics books are always next to each other? *[6 marks]*

4. In a photo there are three families (six Greens, four Browns and seven Grays) arranged in a row. The Browns have had an argument so no Brown will stand next to another Brown. How many different permutations are permitted? *[6 marks]*

5. In a cinema there are 15 seats in a row. In how many ways can seven friends be seated in the same row if

(a) there are no restrictions?

(b) they all want to sit together? *[6 marks]*

6. Five women and four men stand in a line.

(a) In how many arrangements will all four men stand next to each other?

(b) In how many arrangements will all the men stand next to each other and all the women stand next to each other?

(c) In how many arrangements will all the men be apart?

(d) In how many arrangements will all the men be apart and all the women be apart? *[6 marks]*

Summary

- The **product principle** ('AND rule') states the total number of possible outcomes when there are two choices (A and B) to be made is the product of the number of outcomes for each choice:

$$n(A \text{ AND } B) = n(A) \times n(B)$$

- The **addition principle** ('OR rule') states that the number of outcomes when we are interested in either the first choice A *OR* the second choice B being made is the sum of the number of outcomes for each choice. A and B must be mutually exclusive.

$$n(A \text{ OR } B) = n(A) + n(B)$$

- The number of **permutations** (specific arrangements) of n different objects is $n!$; $n! = n\,(n-1)\,(n-2)\ldots \times 2 \times 1$
- The link between one factorial and the next can be shown algebraically as: $(n+1)! = (n+1) \times n!$

- If we are selecting r out of n objects, and the order in which we do so does *not* matter, we describe this as a **combination** and there are $\binom{n}{r}$ or ${}^{n}C_{r}$ ways to do this. This has the formula

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}$$

- There are ${}^{n}P_{r}$ ways of selecting r out of n objects when the order in which we select *does* matter.

$${}^{n}P_{r} = \frac{n!}{(n-r)!}$$

- The **exclusion principle** says that to find the number of permutations which do not have a certain property, find the number of permutations that do have the property and subtract it from the number of all possible permutations.
- If a group of items have to be kept together, treat them as one object. Remember that there may be permutations within this group too.
- If objects have to be kept apart, permute all the other objects and then count the permutations of the 'gaps'.

Introductory problem revisited

If a computer can print a line containing all 26 letters of the alphabet in 0.01 seconds, estimate how long it would take to print all possible permutations of the alphabet.

There are 26! ways to permute all the letters of the alphabet. A calculator tells us that this is approximately 4.03×10^{26} ways. At 0.01 seconds per line, this is 4.03×10^{24} seconds; 4.67×10^{19} days or 1.28×10^{17} years. The age of the universe is estimated at around 14 billion years, so this is approximately 10 million times longer than the universe has existed – it is hard to understand such large numbers!

According to the Guinness World Records™ (Book of Records) the largest number which has ever been found to have a use comes from the field of counting. It is called Graham's Number and it is so large that it is not possible to write it out in normal notation even using all the paper in the world, so a new system of writing numbers had to be invented to describe it.

Mixed examination practice 1

Short questions

1. Seven athletes take part in the 100 m final of the Olympic games. In how many ways can three medals be awarded? *[4 marks]*

2. In how many ways can five different letters be put into five different envelopes? *[5 marks]*

3. In how many ways can ten cartoon characters stand in a queue if Mickey, Bugs Bunny and Jerry must occupy the first three places in some order? *[5 marks]*

4. How many three digit numbers contain no zeros? *[6 marks]*

5. A committee of four children is chosen from eight children. The two oldest children cannot both be chosen. Find the number of ways the committee may be chosen. *[6 marks]*

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6. Solve the equation $(n+1)! = 30(n-1)!$ for $n \in \mathbb{N}$.

(Remember: $\mathbb{N}$ is the set of natural (whole non-negative) numbers.) *[5 marks]*

7. How many permutations of the word 'CAROUSEL' start and end in a consonant? *[5 marks]*

8. Solve the equation $\binom{n}{2} = 105$. *[6 marks]*

9. A group of 15 students contains seven boys and eight girls. In how many ways can a committee of five be selected if it must contain at least one boy? *[6 marks]*

10. Abigail, Bahar, Chris, Dasha, Eustace and Franz are sitting next to each other in six seats in a cinema. Bahar and Eustace cannot sit next to each other. In how many different ways can they permute themselves? *[6 marks]*

11. A committee of five is to be selected from a group of 12 children. The two youngest cannot both be on the committee. In how many ways can the committee be selected? *[6 marks]*

12. A car registration number consists of three different letters followed by five digits chosen from 1–9 (the digits can be repeated). How many different registration numbers can be made? *[6 marks]*

13. A van has eight seats, two at the front, a row of three in the middle and a row of three at the back. If only 5 out of a group of 8 people can drive, in how many different ways can they be arranged in the car? *[6 marks]*

14. Ten people are to travel in one car (taking four people) and one van (taking six people). Only two of the people can drive. In how many ways can they be allocated to the two vehicles? (The permutation of the passengers in each vehicle is not important.) *[7 marks]*

Long questions

1. Five girls, Anya, Beth, Carol, Dasha and Elena, stand in a line. How many possible permutations are there in which

(a) Anya is at one end of the line?

(b) Anya is not at either end?

(c) Anya is at the left of the line or Elena is on the right, or both? *[9 marks]*

2. In how many ways can five different sweets be split amongst two people if

(a) each person must have at least one sweet?

(b) one person can take all of the sweets?

(c) one of the sweets is split into two to be shared, and each person gets two of the remaining sweets? *[9 marks]*

3. In a doctor's waiting room, there are 14 seats in a row. Eight people are waiting to be seen.

(a) In how many ways can they be seated?

(b) Three of the people are all in the same family and they want to sit together. How many ways can this happen?

(c) The family no longer have to sit together, but there is someone with a very bad cough who must sit at least one seat away from anyone else. How many ways can this happen? *[8 marks]*

4. (a) Explain why the number of ways of arranging the letters RRDD, given that all the R's and all the D's are indistinguishable is $\binom{4}{2}$.

(b) How many ways are there of arranging n R's and n D's?

(c) A miner is digging a tunnel on a four by four grid. He starts in the top left box and wants to get to the gold in the bottom right box. He can only tunnel directly right or directly down one box at a time. How many different routes can he take?

(d) What will be the general formula for the number of routes when digging on an n by m grid? *[10 marks]*

5. 12 people need to be split up into teams for a quiz.

(a) Show that the number of ways of splitting them into two groups of the same size is $\frac{1}{2}\binom{12}{6}$.

(b) How many ways are there of splitting them into two groups of any size (but there must be at least one person in each group)?

(c) How many ways are there of splitting them into three groups of four people? *[9 marks]*

6. (a) How many different ways are there to select a group of three from a class of 31 people?

(b) In another class there are 1540 ways of selecting a group of three people. How many people are there in the class?

(c) In another class the teacher noted that the number of ways to select a group of size three is 100 times larger than the number of people in the class. How many people are in the class? *[9 marks]*

3 Polynomials

In this chapter you will learn:

- the formal definition of a polynomial
- operations with polynomials
- a trick for factorising polynomials and finding remainders
- how to sketch the graphs of polynomials
- how to identify the number of solutions of a quadratic equation.

Introductory problem

Without using your calculator, solve the cubic equation

$$x^3 - 13x^2 + 47x - 35 = 0$$

You may think that your calculator can 'magically' do any calculation, such as $\sin 45°$ or e^{-2} or $\ln 2$. However, like any computer, it is built from a device called a logic circuit which can only really do addition. If you repeatedly do addition you get multiplication and if you repeatedly do multiplication you raise to a power. Any expression which only uses these operations is called a polynomial. What is particularly surprising is that all the other operations your calculator does can be approximated very accurately using these polynomials.

3A Working with polynomials

EXAM HINT

The order of a polynomial is sometimes also called its degree.

The **polynomial functions** of x make up a family of functions, each of which can be written as a sum of non-negative integer powers of x. Polynomial functions are classified according to the highest power of x occurring in the function, called the **order of the polynomial**.

General form of the polynomial	Order	Classification	Example
a	0	Constant polynomial	$y = 5$
$ax + b$	1	Linear polynomial	$y = x + 7$
$ax^2 + bx + c$	2	Quadratic polynomial	$y = -3x^2 + 4x - 1$
$ax^3 + bx^2 + cx + d$	3	Cubic polynomial	$y = 2x^3 + 7x$
$ax^4 + bx^3 + cx^2 + dx + e$	4	Quartic polynomial	$y = x^4 - x^3 + 2x + \frac{1}{2}$

The letters a, b, c, … are called the **coefficients** of the powers of x, and the coefficient of the highest power of x in the function (a in the table above) is called the **lead coefficient** and the term containing it is the **leading order term**.

Coefficients can take any value, with the restriction that the lead coefficient cannot equal zero; a polynomial of order n which had a lead coefficient 0 could be more simply written as a polynomial of order $(n-1)$. The sign of the lead coefficient dictates whether the polynomial is a **positive polynomial** $(a>0)$ or a **negative polynomial** $(a<0)$.

Adding and subtracting two polynomials is straightforward – it is just collecting like terms. For example:

$$(x^4+3x^2-1)-(2x^3-x^2+2)=x^4-2x^3+4x^2-3.$$

Multiplying is a little more difficult. Below is one suggested way of setting out polynomial multiplication to ensure that you include all of the terms.

The Greeks knew how to solve quadratic equations, and general cubics and quartics were first solved in 14th Century Italy. For over three hundred years nobody was able to find a general solution to the quintic equation, until in 1821 Niels Abel used a branch of mathematics called group theory to prove that there could never be a 'quintic formula'.

Worked example 3.1

Expand $(x^3+3x^2-2)(x^2-5x+4)$.

Multiply each term in the 1st bracket by the whole of the 2nd bracket

$$
\begin{aligned}
&(x^3+3x^2-2)(x^2-5x+4)\\
&= x^3(x^2-5x+4)+3x^2(x^2-5x+4)-2(x^2-5x+4)\\
&= x^5-5x^4+4x^3\\
&\quad +3x^4-15x^3+12x^2\\
&\quad -2x^2+10x-8\\
&= x^5-2x^4-11x^3+10x^2+10x-8
\end{aligned}
$$

It is important to be able to decide when two polynomials are 'equal'. x^2 and $3x-2$ may take the same value when $x=1$ or 2 but when $x=3$ they do not. However, $3x-2$ and $4x-(x+2)$ are always equal, no matter what value of x you choose.

This leads to what seems like a very obvious definition of equality of polynomials:

You will find that this type of equality is called an identity and is looked at in more detail in Section 4H.

KEY POINT 3.1

Two polynomials being equal means that they have the same order and all of their coefficients are equal.

Ideas similar to comparing coefficients will be applied to vectors (chapter 13) and complex numbers (chapter 15).

However, this definition can be used to solve some quite tricky problems. We tend to use it in questions which start from telling us that two polynomials are equal; we can then **compare coefficients**.

Worked example 3.2

If $(ax)^2 + bx + 3(ax^2 + x) = 4x - 2x^2$ find the values of a and b.

Rearrange each side to make the coefficients clear

LHS: $a^2x^2 + bx + 3ax^2 + 3x$
$= a^2x^2 + 3ax^2 + bx + 3x$
$= x^2(a^2 + 3a) + x(b + 3)$

Compare coefficients and solve the resulting equations

x^2: $(a^2 + 3a) = -2$
$\Leftrightarrow a^2 + 3a + 2 = 0$
$\Leftrightarrow (a + 1)(a + 2) = 0$
$\Leftrightarrow a = -1$ or $a = -2$
x: $b + 3 = 4$
$\Leftrightarrow b = 1$

A very important use of this technique is factorising a polynomial if one factor is already known. To do this we need to know what the remaining factor looks like. For example, if a cubic has one linear factor, the other factor must be quadratic.

Worked example 3.3

$(x-1)$ is a factor of $x^4 + 3x^2 + 2x - 6$. Find the remaining factor.

The remaining factor must be a cubic. We can write the original function as a product of $(x - 1)$ and a general cubic

$$x^4 + 3x^2 + 2x - 6 = (x-1)(ax^3 + bx^2 + cx + d)$$

Multiplying out the right hand side and grouping like terms allows us to compare coefficients

$$= ax^4 + bx^3 + cx^2 + dx$$
$$-ax^3 - bx^2 - cx - d$$
$$= ax^4 + x^3(b-a) + x^2(c-b) + x(d-c) - d$$

continued . . .

Remember that the coefficient of x^3 in the original expression is zero!

Compare x^4:

$a = 1$

Compare x^3:

$b - a = 0$

$b = 1$

Compare x^2:

$c - b = 3$

$c = 4$

Compare x:

$d - c = 2$

$d = 6$

Answer the question

The remaining factor is $x^3 + x^2 + 4x + 6$

'Finding the remaining factor' is another way of asking you to divide. We can conclude from this example that:

$$\frac{x^4 + 3x^2 + 2x - 6}{x - 1} = x^3 + x^2 + 4x + 6$$

EXAM HINT

Notice in the previous example we could check using the constant term that $d = 6$.

Exercise 3A

1. Decide whether each of the following expressions are polynomials. For those that are, give the order and the lead coefficient.

(a) $3x^3 - 3x^2 + 2x$

(b) $1 - 3x - x^5$

(c) $5x^2 - x^{-3}$

(d) $9x^4 - \frac{5}{x}$

(e) $4e^x + 3e^{2x}$

(f) $x^4 + 5x^2 - 3\sqrt{x}$

(g) $4x^5 - 3x^3 + 2x^7 - 4$

(h) 1

2. Expand the brackets for the following expressions:

(a) (i) $(3x - 2)(2x^2 + 4x - 7)$

(ii) $(3x + 1)(x^2 + 5x + 6)$

(b) (i) $(2x + 1)(x^3 - 8x^2 + 6x - 1)$

(ii) $(2x + 5)(x^3 - 6x^2 + 3)$

(c) (i) $(b^2+3b-1)(b^2-2b+4)$

(ii) $(r^2-3r+7)(r^2-8r+2)$

(d) (i) $(5-x^2)(x^4-2x^3+1)$

(ii) $(x-x^3)(x^3-x-1)$

3. Find the remaining factor if

(a) (i) $x^3+3x^2-11x+2$ has a factor of $x-2$

(ii) $x^3+4x^2-3x-18$ has a factor of $x+3$

(b) (i) $6x^3+13x^2+2x-6$ has a factor of $2x+3$

(ii) $25x^3+5x^2-10x-2$ has a factor of $5x+1$

(c) (i) $x^4-5x^3+9x^2-2x-21$ has a factor of $x-3$

(ii) $x^4-5x^3+5x^2+3x-28$ has a factor of $x-4$

(d) (i) $x^4-3x^3+12x^2-15x+35$ has a factor of x^2-3x+7

(ii) x^3-2x^2+3x-6 has a factor of x^2+3

4. Given that the result of each division is a polynomial, simplify each expression.

(a) (i) $\dfrac{x^4-x^3-2x^2+3x-6}{x-2}$ (ii) $\dfrac{x^4+3x^3+2x^2+2x+4}{x+2}$

(b) (i) $\dfrac{2x^4+27x^2+36}{x^2+12}$ (ii) $\dfrac{2x^3-3x^2-27}{2x^2+3x+9}$

5. Find the unknown constants a and b in these identities.

(a) (i) $ax^2+bx=4x^2-6x$ (ii) $ax^2+4=3x^2+4b$

(b) (i) $ax^2+bx=4x+bx^2-ax$ (ii) $ax^2+2bx+6x=0$

(c) (i) $(ax+1)^2+3bx=2ax^2-2x+1$

(ii) $(x+a)^2+b=x^2+4x+9$

(d) (i) $ax^2+bx-2ax=2x-4x^2$

(ii) $ax^2-3bx^2+bx+4x=x^2+7x$

(e) (i) $(ax)^2-(bx)^2+bx=2x$

(ii) $(ax+b)^2=4x^2-20x+25$

6. In what circumstances might you want to expand brackets? In what circumstances is the factorised form better?

7. (a) Is it always true that the sum of a polynomial of order n and a polynomial of order $n-1$ has order n?

(b) Is it always true that the sum of a polynomial of order n and a polynomial of order n has order n?

3B Remainder and factor theorems

We saw in the last section that we can factorise polynomials by comparing coefficients. For example, if we know that $(x+2)$ is one factor of x^3+2x^2+x+2, we can write $x^3+2x^2+x+2=(x+2)(ax^2+bx+c)$ and compare coefficients to find that the other factor is (x^2+1).

If we try to factorise x^3+2x^2+x+5 using $(x+2)$ as one factor, we find that it is not possible; $(x+2)$ is not a factor of x^3+2x^2+x+5. However, using the factorisation of x^3+2x^2+x+2 we can write:

$$x^3+2x^2+x+5=(x+2)(x^2+1)+3$$

The number 3 is the **remainder** – it is what is left over when we try to write x^3+2x^2+x+5 as a multiple of $(x+2)$. In the last section we saw that factorising is related to division. In this case, we could say that:

$$\frac{x^3+2x^2+x+5}{x+2}=(x^2+1)\text{ with remainder }3$$

This is similar to the concept of a remainder when dividing numbers: for example, $25=3\times7+4$, so we would say that 4 is the remainder when 25 is divided by 7.

We can find the reminder by including it as another unknown coefficient. For example, to find the reminder when x^3+2x^2+x+5 is divided by $(x+2)$, we could write

$$x^3+2x^2+x+5=(x+2)(ax^2+bx+c)+R$$

then expand and compare coefficients. This is not a quick task. Luckily there is a shortcut which can help us find the reminder without finding all the other coefficients. If we substitute in a value of x that makes the first bracket equal to zero, in this case $x=-2$, into the above equation, it becomes

$$3=(0)(ax^2+bx+c)+R$$

so $R=3$. This means that R is the value we get when we substitute $x=-2$ into the polynomial expression on the left. Fill-in proof sheet 6 on the CD-ROM, 'Remainder theorem', shows you that the same reasoning can be applied when dividing any polynomial by a linear factor. This leads us to the **Remainder theorem**.

EXAM HINT

Notice that $x = \frac{b}{a}$ is the value which makes $ax - b = 0$.

KEY POINT 3.2

The remainder theorem

The remainder when a polynomial expression is divided by $(ax - b)$ is the value of the expression when $x = \frac{b}{a}$.

Worked example 3.4

Find the remainder when $x^3 + 2x + 7$ is divided by $x + 2$.

Use the remainder theorem by rewriting the divisor in the form $(ax - b)$...

$(x + 2) = (x - (-2))$

... then substitute the value of x (obtained from $x = \frac{b}{a}$) into the expression when $x = \frac{b}{a}$

When $x = -2: (-2)^3 + 2 \times (-2) + 7 = -5$

So the remainder is -5

If the remainder is zero then $(ax - b)$ is a factor. This is summarised by the **factor theorem**.

KEY POINT 3.3

The factor theorem

If the value of a polynomial expression is zero when $x = \frac{b}{a}$, then $(ax - b)$ is a factor of the expression.

Worked example 3.5

Show that $2x - 3$ is a factor of $2x^3 - 13x^2 + 19x - 6$.

To use the factor theorem we need to substitute in $x = \frac{3}{2}$

When $x = \frac{3}{2}$:

$$2 \times \left(\frac{3}{2}\right)^3 - 13\left(\frac{3}{2}\right)^2 + 19 \times \left(\frac{3}{2}\right) - 6$$

$$= \frac{27}{4} - \frac{117}{4} + \frac{57}{2} - 6 = 0$$

continued . . .

Therefore, by the factor theorem, $(2x-3)$ is a factor of $2x^3-13x^2+19x-6$.

We can also use the factor theorem to identify a factor of an expression, by trying several different numbers. Once one factor has been found, then comparing coefficients can be used to find the remaining factors. This is the recommended method for factorising cubic expressions on the non-calculator paper.

Worked example 3.6

Fully factorise $x^3+3x^2-33x-35$.

When factorising a cubic with no obvious factors we must put in some numbers and hope that we can apply the factor theorem

When $x=1$ the expression is -64
When $x=2$ the expression is -81
When $x=-1$ the expression is 0
Therefore $x+1$ is a factor.

We can rewrite the expression as $(x+1)\times$ general quadratic and compare coefficients

$$x^3+3x^2-33x-35$$
$$=(x+1)(ax^2+bx+c)$$
$$=ax^3+(a+b)x^2+(b+c)x+c$$
$$a=1, b=2, c=-35$$
$$x^3+3x^2-33x-35=(x+1)(x^2+2x-35)$$

The remaining quadratic also factorises

$$=(x+1)(x+7)(x-5)$$

EXAM HINT

If the expression is going to factorise easily then you only need to try numbers which are factors of the constant term.

A very common type of question asks you to find unknown coefficients in an expression if factors or remainders are given.

Worked example 3.7

$x^3 + 4x^2 + ax + b$ has a factor of $(x - 1)$ and leaves a remainder of 17 when divided by $(x - 2)$. Find the constants a and b.

Apply factor theorem

when $x = 1$:

$1 + 4 + a + b = 0$

$\Leftrightarrow a + b = -5 \quad (1)$

Apply remainder theorem

when $x = 2$:

$8 + 16 + 2a + b = 17$

$\Leftrightarrow 2a + b = -7 \quad (2)$

Two equations with two unknowns can be solved simultaneously

(2) − (1)

$a = -2$

$b = -3$

Exercise 3B

1. Use the remainder theorem to find the remainder when:
   (a) (i) $x^2 + 3x + 5$ is divided by $x + 1$
   (ii) $x^2 + x - 4$ is divided by $x + 2$
   (b) (i) $x^3 - 6x^2 + 4x + 8$ is divided by $x - 3$
   (ii) $x^3 - 7x^2 + 11x$ is divided by $x - 1$
   (c) (i) $6x^4 + 7x^3 - 5x^2 + 5x + 10$ is divided by $2x + 3$
   (ii) $12x^4 - 10x^3 + 11x^2 - 5$ is divided by $3x - 1$
   (d) (i) x^3 is divided by $x + 2$
   (ii) $3x^4$ is divided by $x - 1$

2. Decide whether each of the following expressions are factors of $2x^3 - 73 - 3x + 2$.
   (a) x
   (b) $x - 1$
   (c) $x + 1$
   (d) $x - 2$
   (e) $x + 2$
   (f) $x - \frac{1}{2}$
   (g) $x + \frac{1}{2}$
   (h) $2x - 1$
   (i) $2x + 1$
   (j) $3x - 1$

3. Fully factorise the following expressions:

(a) (i) x^3+2x^2-x-2 (ii) x^3+x^2-4x-4

(b) (i) $x^3-7x^2+16x-12$ (ii) $x^3+6x^2+12x+8$

(c) (i) $x^3-3x^2+12x-10$ (ii) $x^3-2x^2+2x-15$

(d) (i) $6x^3-11x^2+6x-1$ (ii) $12x^3+13x^2-37x-30$

4. $6x^3+ax^2+bx+8$ has a factor $(x+2)$ and leaves a remainder of -3 when divided by $(x-1)$.
Find a and b. *[5 marks]*

5. x^3+8x^2+ax+b has a factor of $(x-2)$ and leaves a remainder of 15 when divided by $(x-3)$.
Find a and b. *[5 marks]*

6. The polynomial $x^2+kx-8k$ has a factor $(x-k)$. Find the possible values of k. *[5 marks]*

7. The polynomial $x^2-(k+1)x-3$ has a factor $(x-k+1)$. Find k. *[6 marks]*

8. $x^3-ax^2-bx+168$ has factors $(x-7)$ and $(x-3)$.

(a) Find a and b.

(b) Find the remaining factor of the expression. *[6 marks]*

9. x^3+ax^2+9x+b has a factor of $(x-11)$ and leaves a remainder of -52 when divided by $(x+2)$.

(a) Find a and b.

(b) Find the remainder when x^3+ax^2+9x+b is divided by $(x-2)$. *[6 marks]*

10. $f(x)=x^3+ax^2+3x+b$.
The remainder when $f(x)$ is divided by $(x+1)$ is 6. Find the remainder when $f(x)$ is divided by $(x-1)$. *[5 marks]*

$f(x)$ is just a name given to the expression. You will learn more about this notation in chapter 5.

11. The polynomial x^2-5x+6 is a factor of $2x^3-15x^2+ax+b$. Find the values of a and b. *[6 marks]*

3C Sketching polynomial functions

The word quadratic indicates that the term with the highest power in the equation is x^2. It comes from the Latin *quadratus*, meaning square.

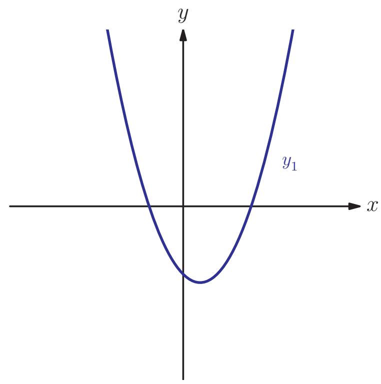
$x \in \mathbb{R}$ means that x can be any number. See Prior learning Section G on the CD-ROM for the meaning of other similar statements.

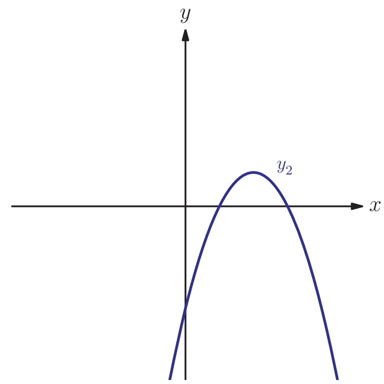
The simplest polynomial with a curved graph is a quadratic.

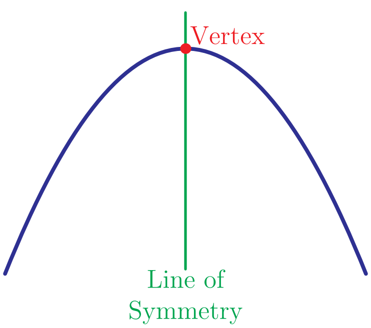
Let us look at two examples of quadratic functions:

$y_1 = 2x^2 + 2x - 4$ and $y_2 = -x^2 + 4x - 3$ $(x \in \mathbb{R})$

You can use your calculator to plot the two graphs. See Calculator sheet 2 on the CD-ROM to see how to sketch graphs on your GDC.

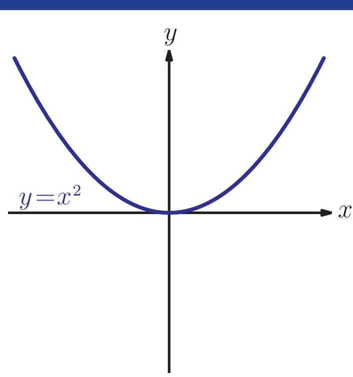
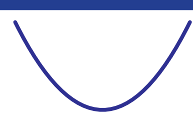
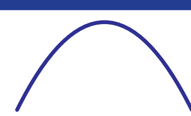


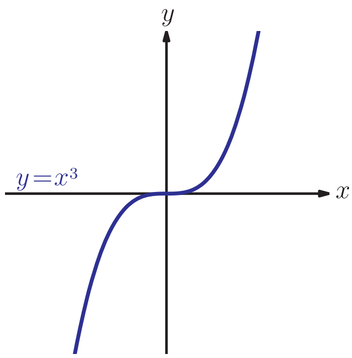
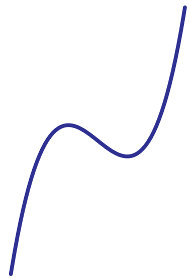
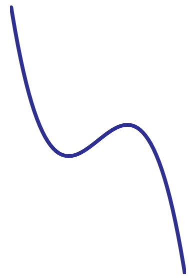
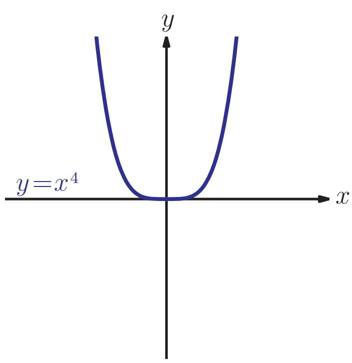
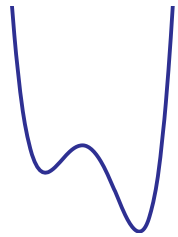
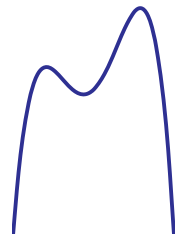




These two graphs have a similar shape, called a **parabola.** A parabola has a single turning point (also called the **vertex**) and a vertical line of symmetry. The most obvious difference is that y_1 has a minimum point, whereas y_2 has a maximum point. This is because of the different signs of the x^2 term.

The graphs of other polynomial functions are also smooth curves. You need to know the shapes to expect for these graphs.

n	$y = x^n$	Positive degree n polynomial	Negative degree n polynomial	Number of times graph can cross or touch the x-axis	Number of turning points
2				0, 1 or 2	1

3				1, 2, or 3	0 or 2
4				0, 1, 2 or 3	1 or 3

The constant term in the polynomial expression gives the position of the y-intercept of the graph (where the curve crosses the y-axis). This is because all other terms contain x, so when $x = 0$ the only non-zero term is the constant term. For example, $2x^3 - 3x + 5 = 5$ when $x = 0$.

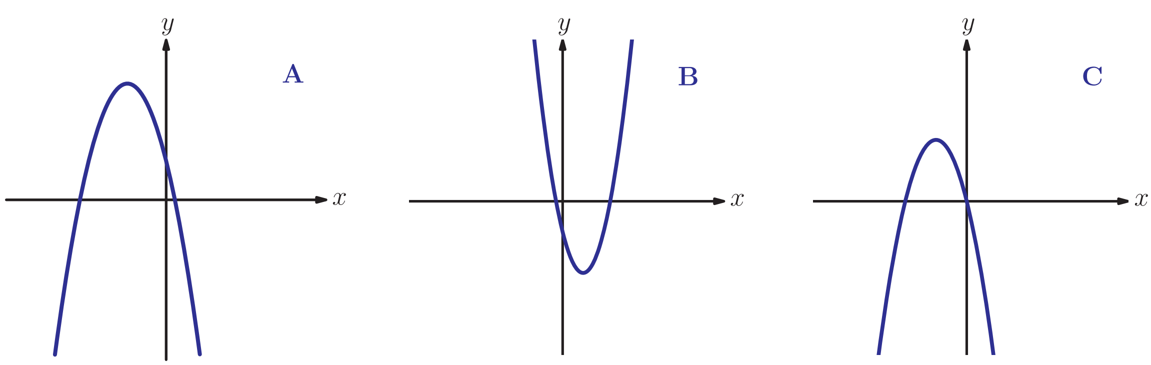
Worked example 3.8

Match each equation to the corresponding graph, explaining your reasons.

(a) $y = 3x^2 - 4x - 1$

(b) $y = -2x^2 - 4x$

(c) $y = -x^2 - 4x + 2$



continued . . .

Graph B is the only positive quadratic

Graph B shows a positive quadratic, so graph B corresponds to equation (a).

We can distinguish between the other two graphs based on their *y*-intercept

Graph A has a positive y-intercept, so graph A corresponds to equation (c).
Graph C corresponds to equation (b).

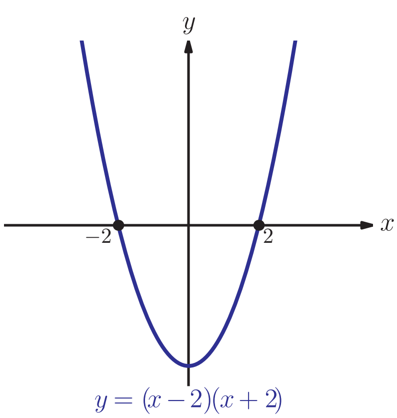
Factorising is covered in Prior learning Section N.

To sketch the graph of a polynomial it is also useful to know its x-intercepts. These are the points where the graph crosses the x-axis, so at those points $y = 0$. For this reason, they are called **zeros** of the polynomial. For example, the quadratic polynomial $x^2 - 5x + 6$ has zeros at $x = 2$ and $x = 3$. They are the **roots** (solutions) of the equation $x^2 - 5x + 6 = 0$ and can be found from the factorised form of the polynomial: $(x-2)(x-3)$.

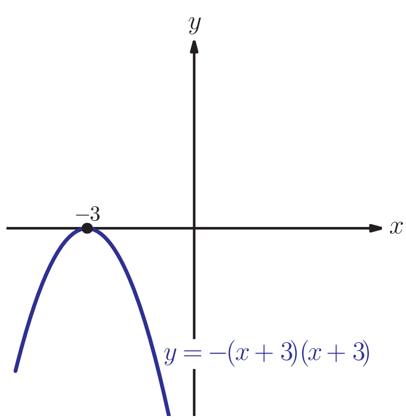
KEY POINT 3.4

The polynomial $a(x-p_1)(x-p_2)(x-p_3)\ldots$ has zeros $p_1, p_2, p_3, \ldots$

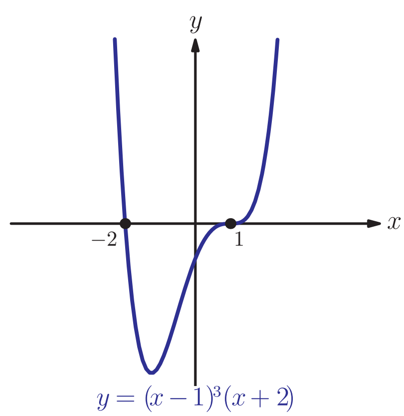
If some of the zeros are the same, we say that the polynomial has a repeated root. For example, the equation $(x-1)^2(x+3)$ has a repeated (double) root $x = 1$ and a single root $x = -3$. Repeated roots tell us how the graph meets the x-axis.



If a polynomial has a factor $(x - a)$ then the curve passes straight through the x-axis at a.



If a polynomial has a double factor $(x - a)^2$ then the curve touches the x-axis at a.



If a polynomial has a triple factor $(x - a)^3$ then the curve passes through the x-axis at a, flattening as it does so.

KEY POINT 3.5

The stages of sketching graphs of polynomials are:

- classify the order of the polynomial and whether it is positive or negative to deduce the basic shape
- set $x = 0$ to find the y-intercept
- write in factorised form
- find x-intercepts
- decide on how the curve meets the x-axis at each intercept
- connect all this information to make a smooth curve.

Worked example 3.9

Sketch the graph of $y = (1-x)(x-2)^2$.

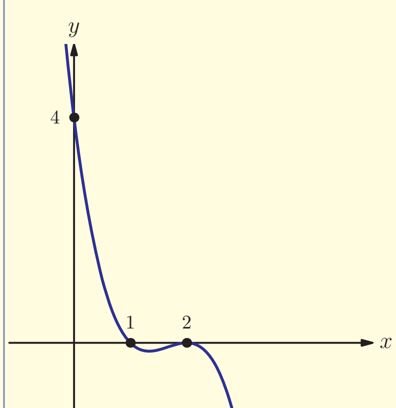
Classify the basic shape — This is a negative cubic

Find y-intercept — When $x = 0$ $\quad y = 1 \times (-2)^2 = 4$

Find x-intercepts — When $y = 0$, $x = 1$ or $x = 2$

Decide on shape of curve at x-intercepts — At $x = 1$ curve passes through the axis. At $x = 2$ curve just touches the axis

Sketch the curve



EXAM HINT

A sketch does not have to be accurate or to scale. It must have approximately the correct shape and all important points, such as axis intercepts, must be clearly labelled.

Sometimes we need to deduce possible equations from a given curve.

We will examine the Fundamental theorem of algebra in more detail in Section 15D.

The first thing we should ask is what order polynomial we need to use. There is a result called the Fundamental Theorem of Algebra, which states that a polynomial of order n can have at most n real roots. So, for example, if the given curve has three x-intercepts, we know that the corresponding polynomial must have degree at least 3. We can then use those intercepts to write down the factors of the polynomial.

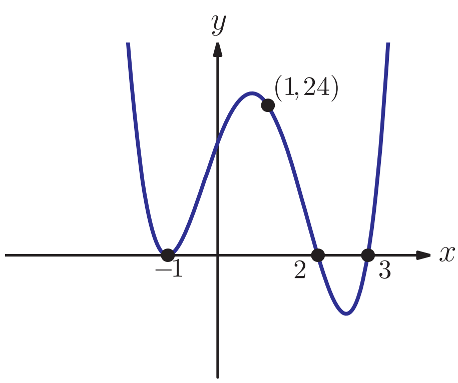
KEY POINT 3.6

To find the equation of a polynomial from its graph:

- use the shape and position of the x-intercepts to write down the factors of the polynomial
- use any other point to find the lead coefficient.

Worked example 3.10

Find a possible equation for this graph.



Describe x-intercepts

Single root at $x = 2$ and $x = 3$
Double root at $x = -1$

Convert this to a factorised form

$$\therefore\ y = k(x+1)^2(x-2)(x-3)$$

Use the fact that when $x = 1$, $y = 24$

$$24 = k \times (2)^2 \times (-1) \times (-2)$$
$$\Leftrightarrow 24 = 8k$$
$$\Leftrightarrow k = 3$$

So the equation is $y = 3(x+1)^2(x-2)(x-3)$

Exercise 3C

1. Sketch the following graphs, labelling all axis intercepts.

(a) (i) $y = 2(x-2)(x-3)$

(ii) $y = 7(x-5)(x+1)$

(b) (i) $y = 4(5-x)(x-3)(x-3)$

(ii) $y = 2(x-1)(2-x)(x-3)$

(c) (i) $y = -(x-4)^2$ (ii) $y = (x-2)^2$

(d) (i) $y = x(x^2+4)$ (ii) $y = (x+1)(x^2-3x+7)$

(e) (i) $y = (1-x)^2(1+x)$ (ii) $y = (2-x)(3-x)^2$

2. Sketch the following graphs, labelling all axis intercepts.

(a) (i) $y = x(x-1)(x-2)(2x-3)$

(ii) $y = (x+2)(x+3)(x-2)(x-3)$

(b) (i) $y = -4(x-3)(x-2)(x+1)(x+3)$

(ii) $y = -5x(x+2)(x-3)(x-4)$

(c) (i) $y = (x-3)^2(x-2)(x-4)$

(ii) $y = -x^2(x-1)(x+2)$

(d) (i) $y = 2(x+1)^3(x-3)$

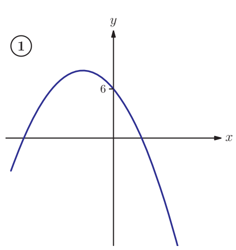
(ii) $y = -x^3(x-4)$

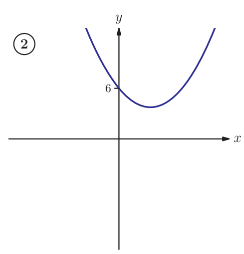
(e) (i) $y = (x^2+3x+12)(x+1)(3x-1)$

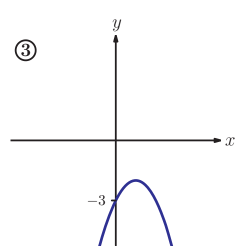
(ii) $y = (x+2)^2(x^2+4)$

3. Match equations and corresponding graphs.

(i) A: $y = -x^2-3x+6$ B: $y = 2x^2-3x+3$ C: $y = x^2-3x+6$

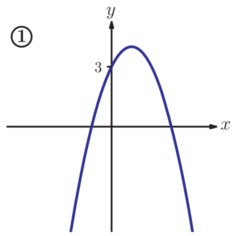






(ii) A: $y = -x^2 + 2x - 3$ B: $y = -x^2 + 2x + 3$ C: $y = x^2 + 2x + 3$

①



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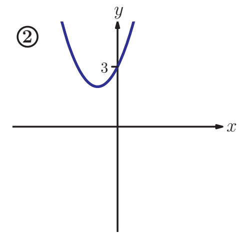

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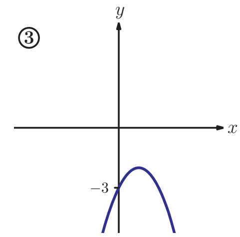
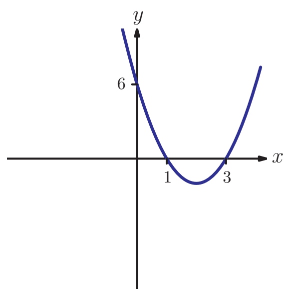
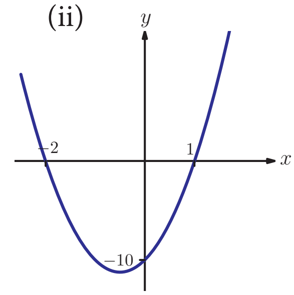


4. The diagrams show graphs of quadratic functions of the form $y = ax^2 + bx + c$. Write down the value of c, and then find the values of a and b.

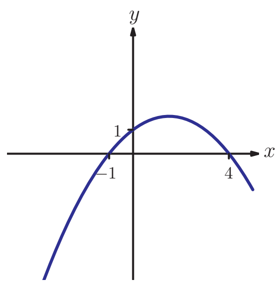
(a) (i)



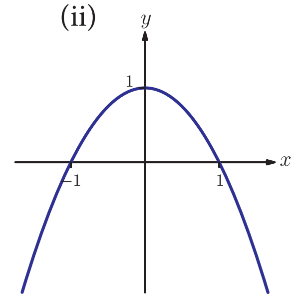
(ii)



(b) (i)

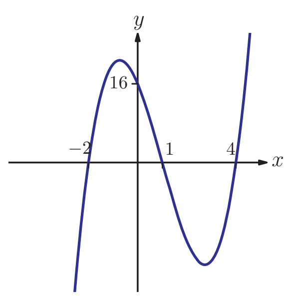


(ii)

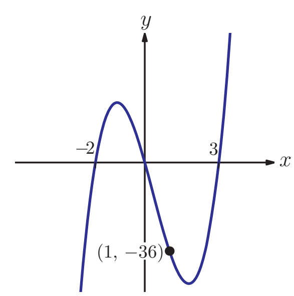


5. Find lowest order polynomial equation for each of these graphs.

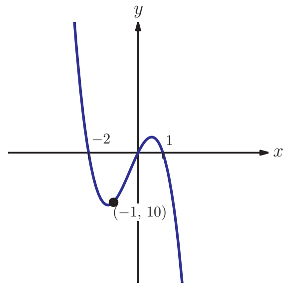
(a) (i)



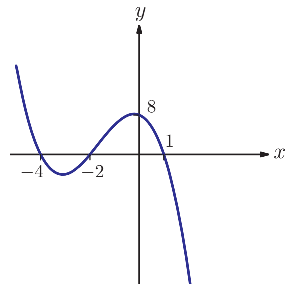
(ii)



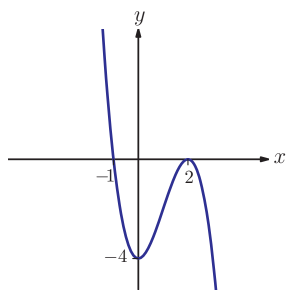
(b) (i)



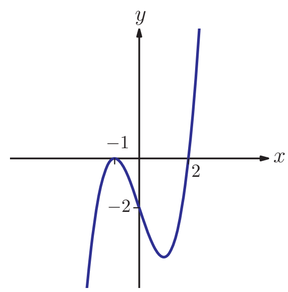
(ii)



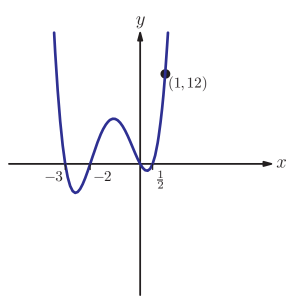
(c) (i)



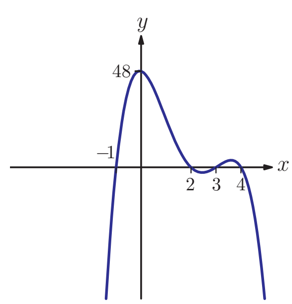
(ii)



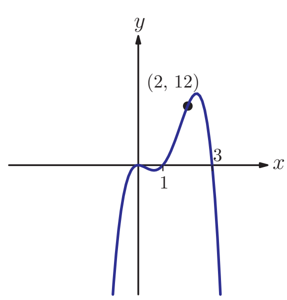
(d) (i)



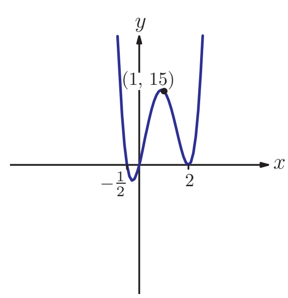
(ii)



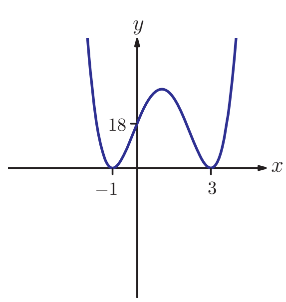
(e) (i)



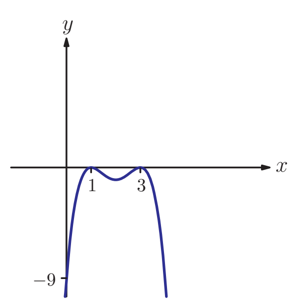
(ii)



(f) (i)



(ii)

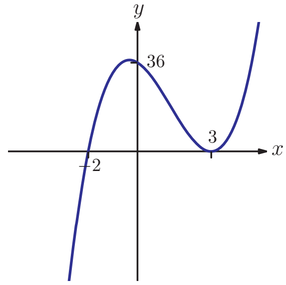


6. (a) Show that $(x-2)$ is a factor of $f(x) = 2x^3 - 5x^2 + x + 2$.
   (b) Factorise $f(x)$.
   (c) Sketch the graph $y = f(x)$.

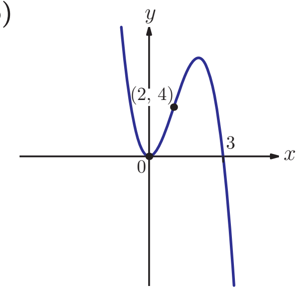
7. Sketch the graph of $y = 2(x+2)^2(3-x)$, labelling clearly any axes intercepts. *[5 marks]*

8. The two graphs below each have equations of the form $y = px^3 + qx^2 + rx + s$. Find the values of p, q, r and s for each graph.

(a)



(b)



[10 marks]

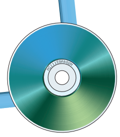
9. (a) Factorise fully $x^4 - q^4$ where q is a positive constant.
   (b) Hence or otherwise sketch the graph $y = x^4 - q^4$, labelling any points where the graph meets an axis. *[5 marks]*

10. (a) Sketch the graph of $y = (x-p)^2(x-q)$ where $p < q$.
   (b) How many solutions does the equation $(x-p)^2(x-q) = k$ have when $k > 0$? *[5 marks]*

EXAM HINT

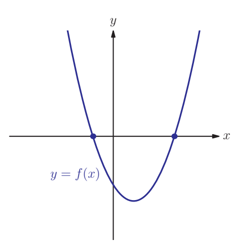
Don't spend too long trying to factorise a quadratic – use the formula in Key point 3.7 if you are asked to find exact solutions and use a calculator (graph or equation solver) otherwise.

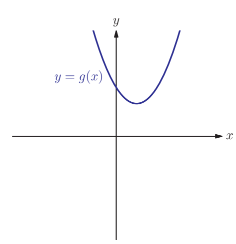
See Calculator sheets 4 and 6 on the CD-ROM.



3D The quadratic formula and the discriminant

It is not always possible to find zeros of a polynomial using factorising. Try factorising the two quadratic expressions $f(x) = x^2 - 3x - 3$ and $g(x) = x^2 - 3x + 3$. It appears that neither of the expressions can be factorised, but sketching the graphs reveals that y_1 has two zeros, while y_2 has no zeros.





In the case where the polynomial is a quadratic we have another option – using the quadratic formula to find the zeros (roots):

KEY POINT 3.7

The zeros of $ax^2 + bx + c$ are given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

You can find the proof of this formula in Fill-in proof 3 'Proving quadratic formula'.

The solution to quadratic equations was known to the Greeks, although they did everything without algebra, using geometry instead. They would have described the quadratic equation $x^2 + 3x = 10$ as 'the area of a square plus three times its length measures ten units'.

Worked example 3.11

Use the quadratic formula to find the zeros of $x^2 - 5x - 3$.

It is not obvious how to factorise the quadratic expression, so use the quadratic formula

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4 \times 1 \times (-3)}}{2}$$

$$= \frac{5 \pm \sqrt{37}}{2}$$

The roots are

$$\frac{5+\sqrt{37}}{2} = 5.54 \text{ and } \frac{5-\sqrt{37}}{2} = -0.541 \text{ (3SF)}$$

EXAM HINT

In International Baccalaureate® exams you should either give exact answers (such as $\frac{5+\sqrt{37}}{2}$) or round your answers to 3 significant figures, unless you are told otherwise. See Prior learning Section B on the CD-ROM for rules of rounding.

Let us examine what happens if we try to apply the quadratic formula to find the zeros of $x^2 - 3x + 3$:

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4 \times 1 \times 3}}{2} = \frac{3 \pm \sqrt{-3}}{2}$$

In chapter 15 you will meet imaginary numbers, a new type of number which makes it possible to find zeros of functions like this.

As the square root of a negative number is not a real number, it follows that the expression has no real zeros.

Looking more closely at the quadratic formula, we see that it can be separated into two parts:

$$x = \frac{-b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

The line of symmetry of the parabola lies halfway between the two roots:

KEY POINT 3.8

The line of symmetry of $y = ax^2 + bx + c$ is

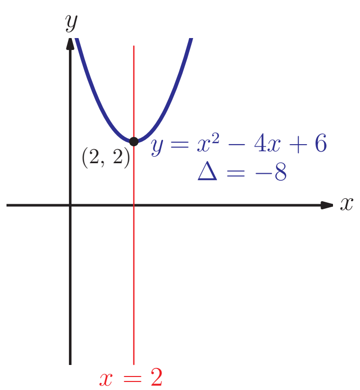
$$x = -\frac{b}{2a}$$

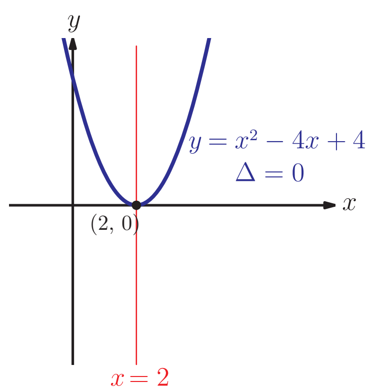
The second part of the formula involves a root expression: $\sqrt{b^2 - 4ac}$.

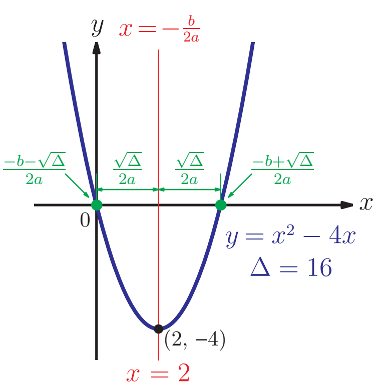
The expression $b^2 - 4ac$ is called the **discriminant** of the quadratic (often symbolised by the Greek letter Δ) and $\frac{\sqrt{b^2 - 4ac}}{2a}$ is the distance of the zeros from the line of symmetry $x = -\frac{b}{2a}$.

The square root of a negative number is not a real number, so if the discriminant is negative, there can be no real zeros and the graph will not cross the x-axis. If the discriminant is zero, the graph is tangent to the x-axis at a point which lies on the line of symmetry (it touches the x-axis rather than crossing it).

The graphs below are examples of the three possible situations:







KEY POINT 3.9

For a quadratic expression $ax^2 + bx + c$, the discriminant is:
$\Delta = b^2 - 4ac$.

- If $\Delta < 0$ the expression has no real zeros
- If $\Delta = 0$ the expression has one (repeated) zero
- If $\Delta > 0$ the expression has two distinct real zeros

There is also a formula for solving cubic equations called 'Cardano's Formula'. It too has a discriminant which can be used to decide how many solutions there will be:

$$b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd$$

This gets a lot more complicated than the quadratic version!

Worked example 3.12

Find the exact values of k for which the quadratic equation $kx^2 - (k+2)x + 3 = 0$ has a repeated root.

Repeated root means that $\Delta = b^2 - 4ac = 0$

$b^2 - 4ac = 0$

We can form an equation in k using $a = k$ and $b = -(k + 2)$

$(k+2)^2 - 4(k)(3) = 0$

This is a quadratic equation

$k^2 + 4k + 4 - 12k = 0$

$k^2 - 8k + 4 = 0$

It doesn't appear to factorise, so use the quadratic formula to find k

$$k = \frac{8 \pm \sqrt{8^2 - 4 \times 4}}{2}$$
$$= \frac{8 \pm \sqrt{48}}{2}$$
$$= \frac{(8 \pm 4\sqrt{3})}{2}$$
$$= 4 \pm 2\sqrt{3}$$

When $\Delta < 0$, the graph does not intersect the x-axis, so it is either entirely above or entirely below it. The two cases are distinguished by the value of a.

Questions involving discriminants often lead to quadratic inequalities, which are covered in Prior learning Section Z.

KEY POINT 3.10

For a quadratic function with $\Delta < 0$:

if $a > 0$ then $y > 0$ for all x

if $a < 0$ then $y < 0$ for all x

Worked example 3.13

Let $y = -3x^2 + kx - 12$.
Find the values of k for which $y < 0$ for all x.

y is a negative quadratic. $y < 0$ means that the graph is entirely below the x-axis. This will happen when $y = 0$ has no real roots

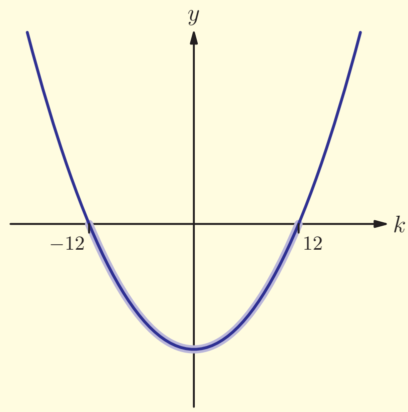
No real roots, $\therefore \Delta < 0$

$k^2 - 4(-3)(-12) < 0$

$k^2 < 144$

This is a quadratic inequality. A sketch of the graph will help

$\Rightarrow k^2 - 144 < 0$



$\therefore -12 < k < 12$

Exercise 3D

1. Evaluate the discriminant of these quadratic expressions.
   (a) (i) $x^2 + 4x - 5$ (ii) $x^2 - 6x - 8$
   (b) (i) $2x^2 + x + 6$ (ii) $3x^2 - x + 10$
   (c) (i) $3x^2 - 6x + 3$ (ii) $9x^2 - 6x + 1$
   (d) (i) $12 - x - x^2$ (ii) $-x^2 - 3x + 10$

2. State the number of zeros for each expression in Question 1.

3. Find the exact solutions of these equations.

(a) (i) $x^2 - 3x + 1 = 0$ (ii) $x^2 - x - 1 = 0$

(b) (i) $3x^2 + x - 2 = 0$ (ii) $2x^2 - 6x + 1 = 0$

(c) (i) $4 + x - 3x^2 = 0$ (ii) $1 - x - 2x^2 = 0$

(d) (i) $x^2 - 3 = 4x$ (ii) $3 - x = 2x^2$

Some of the many applications of quadratic equations are explored in Supplementary sheet 1 'The many applications of quadratic equations'.

4. Find the values of k for which:

(a) (i) the equation $x^2 - x + k =$ has two distinct real roots

(ii) the equation $3x^2 - 5x + k = 0$ has two distinct real roots

(b) (i) the equation $5x^2 - 2x + (2k - 1) = 0$ has equal roots

(ii) the equation $2x^2 + 3x - (3k + 1) = 0$ has equal roots

(c) (i) the equation $-x^2 + 3x + (k - 1) = 0$ has real roots

(ii) the equation $-2x^2 + 3x - (2k + 1) = 0$ has real roots

(d) (i) the equation $3kx^2 - 3x + 2 = 0$ has no real solutions

(ii) the equation $kx^2 + 5x + 3 = 0$ has no real solutions

(e) (i) the quadratic expression $(k - 2)x^2 + 3x + 1$ has a repeated zero

(ii) the quadratic expression $-4x^2 + 5x + (2k - 5)$ has a repeated zero

(f) (i) the graph of $y = x^2 - 4x + (3k + 1)$ is tangent to the x-axis

(ii) the graph of $y = -2kx^2 + x - 4$ is tangent to the x-axis

(g) (i) the expression $-3x^2 + 5k$ has no real zeros

(ii) the expression $2kx^2 - 3$ has no real zeros

5. Find the values of parameter m for which the quadratic equation $mx^2 - 4x + 2m = 0$ has equal roots. *[5 marks]*

6. Find the exact values of k such that the equation $-3x^2 + (2k + 1)x - 4k = 0$ has a repeated root. *[6 marks]*

7. Find the range of values of the parameter c such that $2x^2 - 3x + (2c - 1) \geq 0$ for all x. *[6 marks]*

8. Find the set of values of k for which the equation $x^2 - 2kx + 6k = 0$ has no real solutions. *[6 marks]*

9. Find the range of value of k for which the quadratic equation $kx^2 - (k + 3)x - 1 = 0$ has no real roots. *[6 marks]*

10. Find the range of values of m for which the equation $mx^2 + mx - 2 = 0$ has one or two real roots. *[6 marks]*

11. Find the possible values of m such that $mx^2 + 3x - 4 < 0$ for all x. *[6 marks]*

12. The positive difference between the zeros of the quadratic expression $x^2 + kx + 3$ is $\sqrt{69}$. Find the possible values of k. *[5 marks]*

Summary

- **Polynomials** are expressions involving only addition, multiplication and raising to a power.
- If two polynomials are equal we can **compare coefficients**. This can be used to divide two polynomials.
- The **remainder theorem** says that the remainder when a polynomial is divided by $ax - b$ is the value of the polynomial expression when $x = \frac{b}{a}$.
- The **factor theorem** says that if a polynomial expression is zero when $x = \frac{b}{a}$ then $ax - b$ is a factor of the expression.
- The graphs of polynomial are best sketched using their factorised form. Look for repeated factors and find where the graph crosses the axes.
- You can use the shape and position of the x-intercepts to find the equation of a polynomial graph.
- The number of solutions of a quadratic equation depends on the value of the **discriminant**, $\Delta = b^2 - 4ac$:
  - If $\Delta > 0$ there are two distinct solutions.
  - If $\Delta = 0$ there is one solution.
  - If $\Delta < 0$ there is no solution.

Introductory problem revisited

Without using your calculator, solve the cubic equation

$$x^3 - 13x^2 + 47x - 35 = 0$$

If we put $x = 1$ into this expression we get 0, so we can conclude that $x - 1$ is a factor. The remaining quadratic factor can be found by comparing coefficients; it is $x^2 - 12x + 35$, which factorises to give $(x - 5)(x - 7)$.

The equation therefore becomes $(x - 1)(x - 5)(x - 7) = 0$ which has solutions $x = 1, 5,$ or 7.

Mixed examination practice 3

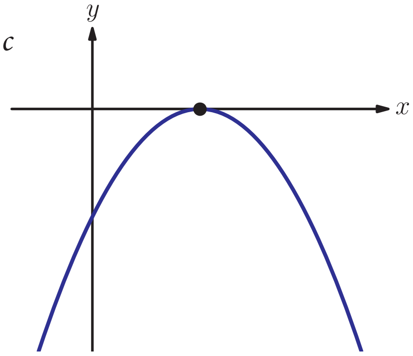
Short questions

1. A quadratic graph passes through the points $(k, 0)$ and $(k+4, 0)$. Find in terms of k the x-coordinates of the turning point. *[4 marks]*

2. The diagram shows the graph of $y = ax^2 + bx + c$

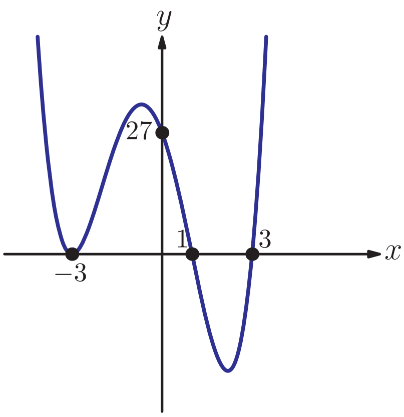
Complete the table to show whether each expression is positive, negative or zero.

expression	positive	negative	zero
a			
c			
$b^2 - 4ac$			
b			



[6 marks]
(© *IB Organization 2000*)

3. The diagram shows the graph with equation $y = ax^4 + bx^3 + cx^2 + dx + e$.
Find the values of a, b, c, d and e. *[6 marks]*



4. The remainder when $(ax+b)^3$ is divided by $(x-2)$ is 8 and the remainder when it is divided by $(x+3)$ is -27. Find the values of a and b. *[5 marks]*

5. (a) Show that $(x-2)$ is a factor of $f(x) = x^3 - 4x^2 + x + 6$.
(b) Factorise $f(x)$.
(c) Sketch the graph of $y = f(x)$. *[7 marks]*

6. The remainder when $(ax+b)^4$ is divided by $(x-2)$ is 16 and the remainder when it is divided by $(x+1)$ is 81. Find the possible values of a and b. *[6 marks]*

7. Sketch the graph of $y=(x-a)^2(x-b)(x-c)$ where $b<0<a<c$. *[5 marks]*

8. Find the exact values of k for which the equation $2kx^2+(k+1)x+1=0$ has equal roots. *[5 marks]*

9. Find the set of values of k for which the equation $2x^2+kx+6=0$ has no real roots. *[6 marks]*

10. Find the range of values of k for which the quadratic function $x^2-(2k+1)x+5$ has at least one real zero. *[6 marks]*

11. The polynomial x^2-4x+3 is a factor of the polynomial $x^3+ax^2+27x+b$. Find the values of a and b. *[6 marks]*

12. Let α and β denote the roots of the quadratic equation

$x^2-kx+(k-1)=0$.

(a) Express α and β in terms of the real parameter k.

(b) Given that $\alpha^2+\beta^2=17$, find the possible values of k. *[7 marks]*

13. Let $q(x)=kx^2+(k-2)x-2$. Show that the equation $q(x)=0$ has real roots for all values of k. *[7 marks]*

14. Find the range of values of k such that for all x, $kx-2\le x^2$. *[7 marks]*

Long questions

1. (a) Find the coordinates of the point where the curve $y=x^2+bx-a$ crosses the y-axis, giving your answer in terms of a and/or b.

(b) State the equation of the axis of symmetry of x^2+bx-a, giving your answer in terms of a and/or b.

(c) Show that the remainder when x^2+bx-a is divided by $x-\frac{a}{b}$ is always positive.

(d) The remainder when x^2+bx-a is divided by $x-a$ is -9. Find the possible values that b can take.

[14 marks]

2. (a) Show that for all values of p, $(x-2)$ is a factor of

$$f(x) = x^3 + (p-2)x^2 + (5-2p)x - 10.$$

(b) By factorising $f(x)$, or otherwise, find the exact values of p for which the equation

$$x^3 + (p-2)x^2 + (5-2p)x - 10 = 0$$

has exactly two real roots.

(c) For the smaller of the two values of p found above, sketch the graph of $y = f(x)$.

[10 marks]

3. (a) On the graph of $y = \dfrac{x^2+4x+5}{x+2}$ prove that there is no value of x for which $y = 0$.

(b) Find the equation of the vertical asymptote of the graph.

(c) Rearrange the equation to find x in terms of y.

(d) Hence show that y cannot take values between -2 and 2.

(e) Sketch the graph of $y = \dfrac{x^2+4x+5}{x+2}$.

[18 marks]

4. Let $f(x) = x^4 + x^3 + x^2 + x + 1$.

(a) Evaluate $f(1)$.

(b) Show that $(x-1)f(x) \equiv x^5 - 1$.

(c) Sketch $y = x^5 - 1$.

(d) Hence show that $f(x)$ has no real roots.

[10 marks]

14. A piecewise function is defined by:

$$f(x)=\begin{cases}2+(x-1)^2, & x<1\\ k-(x-1)^2, & x\ge 1\end{cases}$$

(a) Find the largest value of k for which f is a one-to-one function.

(b) When $k=0$:

(i) Find the range of f.

(ii) Find an expression for $f^{-1}(x)$. *[8 marks]*

15. A function is called *self-inverse* if $f(x)=f^{-1}(x)$ for all x in the domain.

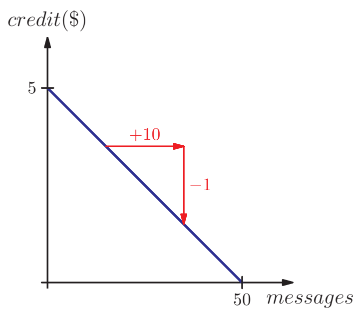
(a) Show that $f(x)=\dfrac{1}{x}$, $x\ne 0$ is a self-inverse function.

(b) Find the value of the constant k so that

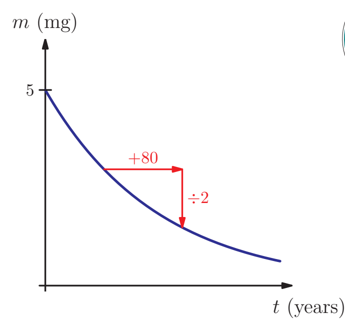
$g(x)=\dfrac{3x-5}{x+k}$, $x\ne k$ is a self-inverse function. *[8 marks]*

5F Rational functions

There are many situations where one quantity deceases as another increases. For example, the amount of your phone credit decreases as the number of text messages you send increases. As the number of messages increases by a fixed number, the credit decreases by a fixed amount; this is called **linear decay**. Another example is radioactive decay, where the amount of a radioactive substance halves in a fixed time period (called half-life). So as the time increases by a fixed number, the amount of substance decreases by a fixed factor; this is called **exponential decay**.



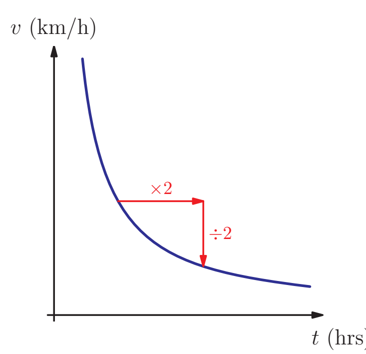
linear decay



exponential decay

Exponential functions are covered in chapter 2, and linear functions in Section R of Prior learning.

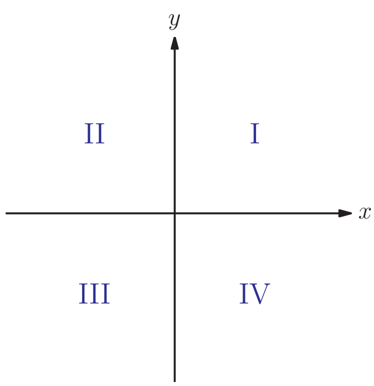
In this chapter we will look at a third type of decay, called **inverse proportion**, where as one quantity increases by a fixed factor, another decreases by the same factor. For example, if you double your speed, the amount of time to travel a given distance will halve. If the total distance travelled is 12 km, then the equation for time (in hours) in terms of speed (in km/h) is $t=12/v$. This is an example of a **reciprocal function**.



EXAM HINT

The reciprocal of a real number $x \neq 0$ is $\frac{1}{x}$. For example, the reciprocal of -2 is $-\frac{1}{2}$ and the reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$.

Graphs of reciprocal functions all have the same shape, called a **hyperbola**. This curve consists of two parts and has the axis as the asymptotes. This means that the function is not defined for $x = 0$, and that as x gets very large (positive or negative), y approaches zero. Therefore neither x nor y can equal zero. The two parts of the curve can be either in the first and third, or the second and fourth quadrants, depending on the sign of k.



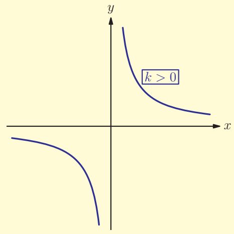
Quadrants

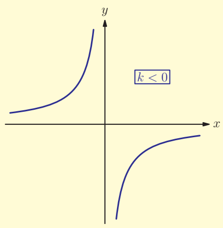
KEY POINT 5.11

A reciprocal function has the form $f(x) = \frac{k}{x}$.

The domain of f is $x \neq 0$ and the range is $y \neq 0$.

The graph of $f(x)$ is a hyperbola.

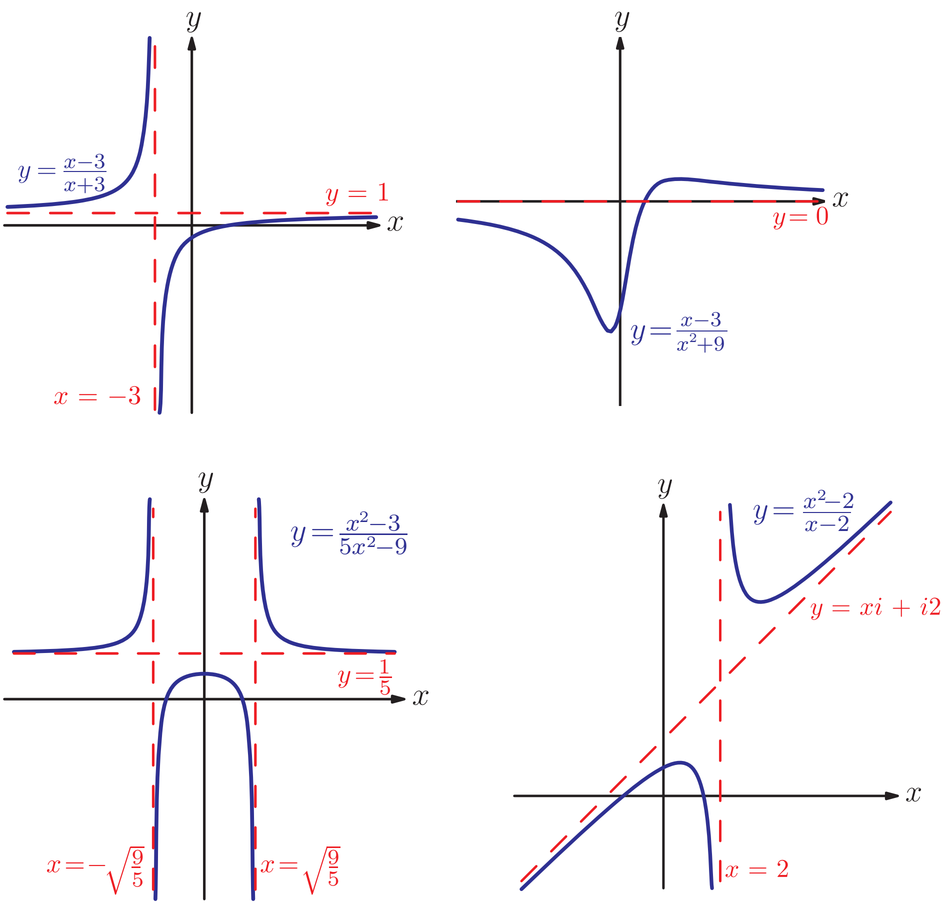




Related to reciprocal functions, **rational functions** are a ratio of two polynomials: $f(x) = \dfrac{p(x)}{q(x)}$.

Rational functions can take many different shapes, depending on the polynomials $p(x)$ and $q(x)$, their orders and how many zeros they have.

Here are some examples of the types of graphs we can get:



One very common feature of rational functions is asymptotes. Three of the four graphs above have **vertical asymptotes**, corresponding to the values of x where the rational function is not defined because of division by zero. The first three graphs have **horizontal asymptotes**, corresponding to the value of y the function approaches as x gets very large.

We have already encountered asymptotes in Sections 2B and 2F.

If the order of the numerator is greater than the order of the denominator then, as x gets larger, the graph gets closer and closer to a polynomial curve.

Understanding this asymptotic behaviour is important in applications which deal with very large numbers (such as in astronomy), but it is also used in some more unexpected situations, for example in looking for prime numbers.

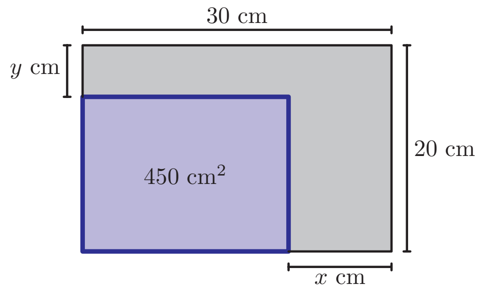
We will only consider rational functions of the form $f(x)=\frac{ax+b}{cx+d}$. The following example illustrates one application of such functions.

Worked example 5.14

A rectangular piece of card has dimensions 30 cm by 20 cm. Strips of width x cm and y cm are cut off the ends, so that the remaining card has area 450 cm^2.

(a) Find an expression for y in terms of x in the form $y=\frac{ax-b}{cx-d}$.

(b) Sketch the graph of y against x.



Write the equation for the remaining area in terms of x and y

(a) $(30-x)(20-y)=450$

We want to make y the subject, so divide by $(30-x)$ rather than expanding the brackets

$$\Leftrightarrow 20-y=\frac{450}{30-x}$$

$$\Leftrightarrow \quad y=20-\frac{450}{30-x}$$

$$=\frac{20(30-x)-450}{30-x}$$

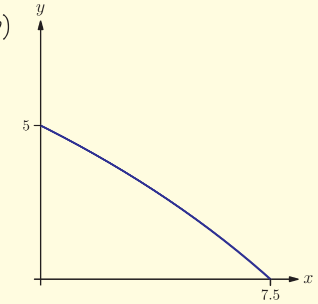
$$=\frac{150-20x}{30-x}$$

We can multiply top and bottom by -1

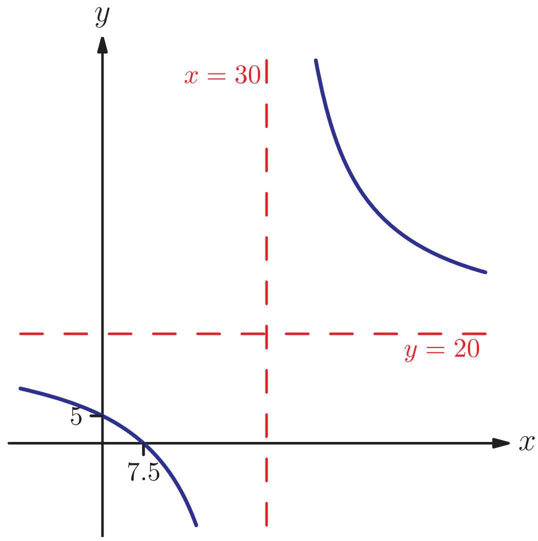
$$\Leftrightarrow y=\frac{20x-150}{x-30}$$

We can use GDC to sketch the graph
Only positive values of x and y are relevant

(b)



In the above example only positive values of x and y were relevant, but we can look at the function $f(x)=\frac{20x-150}{x-30}$ over the whole of $\mathbb{R}$. The function is not defined for $x=30$ (as we would be dividing by zero), so there is a vertical asymptote there. The y-intercept is $(0,5)$. We can find the x-intercept by setting the top of the fraction equal to zero – it is $(7.5,0)$. We can use the calculator to sketch the graph. The curve looks like a hyperbola with the horizontal asymptote $y=20$.



The position of the horizontal asymptote can be found by looking at the first equation we found for y in the worked example, which can also be written as $y=20+\frac{450}{x-30}$. As x gets very large (either positive or negative), $x-30$ gets very large, so $\frac{450}{x-30}$ gets very small. Therefore the value of y gets closer and closer to 20. Another way to find the asymptote is to think about what happens as x gets large in the equation $y=\frac{20x-150}{x-30}$. The terms containing x become much larger than 150 and 30, so the two constant terms can be ignored, leaving $y\approx\frac{20x}{x}=20$.

The above example illustrates all the important properties of rational functions of the form $f(x)=\frac{ax+b}{cx+d}$.

Is zero the same as nothing? What happens if you say that the result of dividing by zero is infinity? What would $\frac{0}{0}$ be?

EXAM HINT

In the examination you can just use the result that the horizontal asymptote is found by dividing the two coefficients of x.

EXAM HINT

Make sure you include all asymptotes and intercepts when sketching graphs of rational functions. The intercepts should help you decide which quadrants the graph is in.

KEY POINT 5.12

The graph of a **rational function** of the form $f(x)=\dfrac{ax+b}{cx+d}$ is a hyperbola with:

- vertical asymptote $x=-\dfrac{d}{c}$ (this is where $cx+d=0$)
- horizontal asymptote $y=\dfrac{a}{c}$
- x-intercept at $x=-\dfrac{b}{a}$ (this is where $ax+b=0$)
- y-intercept at $y=\dfrac{b}{d}$ (this is where $x=0$)

Knowing the position of the asymptotes gives you the domain and range of the function.

Worked example 5.15

Find the domain and range of the function $f(x)=\dfrac{3x-4}{2x+1}$.

The only value excluded from the domain is when the denominator is zero

$2x+1=0$

$\Rightarrow x=-\dfrac{1}{2}$

The domain is:

$x\in\mathbb{R}, x\neq-\dfrac{1}{2}$

Sketching the graph can show us the range. We need to find the horizontal asymptote: divide the coefficients of x

Horizontal asymptote:

$y=\dfrac{3}{2}$

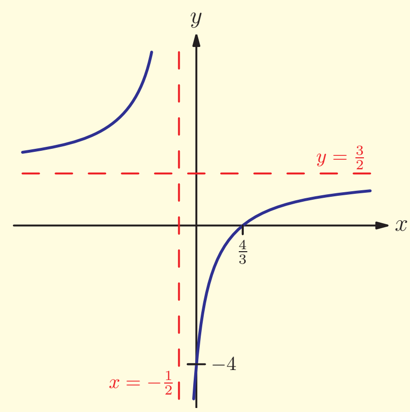
Find the intercepts to decide which quadrants the graph is in

Intercepts:

$x=0\Rightarrow y=-4$

$y=0\Rightarrow 3x-4=0\Rightarrow x=\dfrac{4}{3}$

continued . . .



The range is:

$y \in \mathbb{R}, y \neq \dfrac{3}{2}$

Is there any point learning how to sketch graphs when you have a graphical calculator (aside from the fact that there is a non-calculator paper!)? Do you sometimes need knowledge in order to acquire new knowledge?

Exercise 5F

1. Find the coordinates of all axis intercepts of the following rational functions:

(a) (i) $f(x) = \dfrac{3x+1}{x+3}$ (ii) $f(x) = \dfrac{2x+5}{x+1}$

(b) (i) $f(x) = \dfrac{2x-3}{2x+7}$ (ii) $f(x) = \dfrac{3x-5}{x+2}$

2. Find the equations of the asymptotes of the following graphs:

(a) (i) $y = \dfrac{4x+3}{x-1}$ (ii) $y = \dfrac{2x+1}{x-7}$

(b) (i) $y = \dfrac{3x+2}{2x-1}$ (ii) $y = \dfrac{4x+1}{3x-5}$

(c) (i) $y = \dfrac{3-x}{2x+5}$ (ii) $y = \dfrac{2x+1}{2-3x}$

(d) (i) $y = \dfrac{3}{x-2}$ (ii) $y = \dfrac{2}{2x+1}$

3. Sketch the graphs of the following rational functions, labelling all the axis intercepts and asymptotes.

(a) (i) $y=\frac{2x+1}{x-2}$ (ii) $y=\frac{3x+1}{x-3}$

(b) (i) $y=\frac{x-3}{4-x}$ (ii) $y=\frac{5-x}{x-2}$

(c) (i) $y=\frac{2}{x+3}$ (ii) $y=\frac{1}{x-2}$

(d) (i) $y=-\frac{3}{x}$ (ii) $y=-\frac{2}{x}$

4. Find the domain, range and the inverse function of the following rational functions:

(a) (i) $f(x)=\frac{3}{x}$ (ii) $f(x)=\frac{7}{x}$

(b) (i) $f(x)=\frac{2}{x-3}$ (ii) $f(x)=\frac{5}{x+1}$

(c) (i) $f(x)=\frac{2x+1}{3x-1}$ (ii) $f(x)=\frac{4x-5}{2x+1}$

(d) (i) $f(x)=\frac{5-2x}{x+2}$ (ii) $f(x)=\frac{3x-1}{4x-3}$

5. Find the equations of the asymptotes of the graph of $y=\frac{3x-1}{4-5x}$. *[3 marks]*

6. (a) Find the domain and range of the function $f(x)=\frac{1}{x+3}$.

(b) Find $f^{-1}(x)$. *[5 marks]*

7. Sketch the graph of $y=\frac{3x-1}{x-5}$. *[5 marks]*

8. A function is defined by $f:x\mapsto\frac{ax+3}{2x-8}, x\neq 4$, where $a\in\mathbb{R}$.

(a) Find, in terms of *a*, the range of *f*.

(b) Find the inverse function $f^{-1}(x)$.

(c) Find the value of *a* such that *f* is a self-inverse function. *[5 marks]*

Summary

- In this chapter we have looked at the difference between a function and a relation.
  - a **relation** is any set of paired inputs and outputs, while a **function** is a particular sort of relation where for each input there is only one output
  - we can distinguish between these using the **vertical line test**.
- We met two different types of function.
  - in a **one-to-one function**, each output comes from a unique input
  - in a **many-to-one function**, some outputs may come from more than one input
  - we distinguish between these using the **horizontal line test**.
- To fully define a function, as well as stating a rule, we also need to know its **domain** – the set of allowed inputs. Once we know the domain we can also find the range – the set of outputs resulting from that domain.
- A function can act upon the output of another function. The effect is called a **composite function** and we write $f(g(x))$ or $fg(x)$ or $f \circ g(x)$ for g followed by f.
- Reversing the effect of a function is achieved by applying an **inverse function**, $f^{-1}(x)$.
- Only one-to-one functions have inverse functions. The general method to find an inverse function is:
  - start with $y = f(x)$
  - rearrange to get x in terms of y
  - replace each appearance of y with an x.
- You need to know these properties of **inverse functions**
  - the graph of the inverse function is a reflection in the line $y = x$ of the graph of the original function
  - the domain of the inverse function is the range of the original function, and the range of the inverse function is the domain of the original function
  - composing a function with its inverse gives the identity function:

$$f\left(f^{-1}(x)\right) = f^{-1}\left(f(x)\right) = x$$

- A **self-inverse function** has $f^{-1}(x) = f(x)$. This is equivalent to $f\left(f(x)\right) = x$.

- Rational functions of the form $f(x)=\frac{ax+b}{cx+d}$ have a graph called a **hyperbola** with the following properties:
  - vertical asymptote $x=-\frac{d}{c}$
  - horizontal asymptote $y=\frac{a}{c}$
  - x-intercept at $x=-\frac{b}{a}$
  - y-intercept at $y=\frac{b}{d}$
- A special case of a rational function is a **reciprocal function**, $f(x)=\frac{k}{x}$, which is an example of a self-inverse function.

Introductory problem revisited

Think of any number. Add 3. Double your answer. Take away 6. Divide by the number you first thought of. Is your answer always prime? Why?

We can write the introductory problem as a function:

$$f(x)=\frac{2(x+3)-6}{x}$$

In most cases this simplifies to give 2, which is a prime number.

However, we should now be used to the idea that a function is about more than just a rule, it also needs a domain. The domain for this function cannot include zero, so it gives a prime number for any input other than zero, when the process fails.

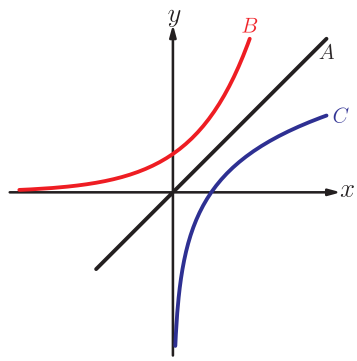
Mixed examination practice 5

Short questions

1. Find the inverses of the following functions:

(a) $f(x)=\log_3(x+3),\ x>0$ (b) $g(x)=3e^{x^3-1}$ *[5 marks]*

2. The diagram shows three graphs.



A is part of the graph of $y = x$.
B is part of the graph of $y = 2^x$.
C is the reflection of graph B in line A.
Write down:

(a) The equation of C in the form $y = f(x)$

(b) The coordinates of the point where C cuts the x-axis. *[5 marks]*

3. (a) Write down the equations of all asymptotes of the graph of $y=\dfrac{4x-3}{5-x}$.

(b) Find the inverse function of $f(x)=\dfrac{4x-3}{5-x}$. *[6 marks]*

4. The function f is given by $f(x)=x^2-6x+10$, for $x\geq 3$.

(a) Write $f(x)$ in the form $(x-p)^2+q$.

(b) Find the inverse function $f^{-1}(x)$.

(c) State the domain of $f^{-1}(x)$. *[6 marks]*

5. If $h(x)=x^2-6x+2$:

(a) Write $h(x)$ in the form $(x-p)^2+q$.

(b) Hence or otherwise find the range of $h(x)$.

(c) By using the largest possible domain of the form $x>k$ where, find the inverse function $h^{-1}(x)$. *[7 marks]*

6. The function $f(x)$ is defined by $f(x)=\dfrac{3-x}{x+1}, x\neq -1$.

(a) Find the range of f.

(b) Sketch the graph of $y=f(x)$.

(c) Find the inverse function of f in the form $f^{-1}(x)=\dfrac{ax+b}{cx+d}$. State its domain and range. *[11 marks]*

7. A function is defined by:

$$f(x)=\begin{cases}5-x, & x<0\\ pe^{-x}, & x\geq 0\end{cases}$$

(a) Given that $p=3$,

(i) Find the range of $f(x)$.

(ii) Find an expression for $f^{-1}(x)$ and state its domain.

(b) Find the value of p for which $f(x)$ is continuous. *[7 marks]*

8. The functions $f(x)$ and $g(x)$ are given by $f(x)=\sqrt{x-2}$ and $g(x)=x^2+x$. The function $f\circ g(x)$ is defined for $x\in\mathbb{R}$ except for the interval $]\,a,b[$.

(a) Calculate the value of a and of b.

(b) Find the range of $f\circ g$. *[7 marks]*

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Long questions

1. If $f(x)=x^2+1,\ x>3$ and $g(x)=5-x$:

(a) evaluate $f(3)$.

(b) Find and simplify an expression for $gf(x)$.

(c) State the geometric relationship between the graphs of $y=f(x)$ and $y=f^{-1}(x)$.

(d) (i) Find an expression for $f^{-1}(x)$.

(ii) Find the range of $f^{-1}(x)$.

(iii) Find the domain of $f^{-1}(x)$.

(e) Solve the equation $f(x)=g(3x)$. *[10 marks]*

2. If $f(x)=2x+1$ and $g(x)=\dfrac{x+3}{x-1},\ x\neq 1$

(a) find and simplify

(i) $f(7)$ (ii) the range of $f(x)$

(iii) $fg(x)$ (iv) $ff(x)$

(b) Explain why $gf(x)$ does not exist.

(c) (i) Find the form of $g^{-1}(x)$.

(ii) State the domain of $g^{-1}(x)$.

(iii) State the range of $g^{-1}(x)$. *[9 marks]*

3. The functions f and g are defined over the domain of all real numbers, $g(x) = e^x$.

(a) Write $f(x) = x^2 + 4x + 9 \;\; x \in \mathbb{R}$ in the form $f(x) = (x+p)^2 + q$.

(b) Hence sketch the graph of $y = x^2 + 4x + 9$, labelling carefully all axes intercepts and the coordinates of the turning point.

(c) State the range of $f(x)$ and $g(x)$.

(d) Hence or otherwise find the range of $h(x) = e^{2x} + 4e^x + 9$. *[10 marks]*

4. Given that $(2x+3)(4-y) = 12$ for $x, y \in \mathbb{R}$:

(a) Write y in terms of x, giving your answer in the form $y = \dfrac{ax+b}{cx+d}$.

(b) Sketch the graph of y against x.

(c) Let $g(x) = 2x + k$ and $h(x) = \dfrac{8x}{2x+3}$.

(i) Find $h(g(x))$.

(ii) Write down the equations of the asymptotes of the graph of $y = h(g(x))$.

(iii) Show that when $k = -\dfrac{19}{2}$, $h(g(x))$ is a self-inverse function. *[17 marks]*

5. (a) Show that if $g(x) = \dfrac{1}{x}$ then $gg(x) = x$.

(b) A function satisfies the identity $f(x) + 2f\left(\dfrac{1}{x}\right) = 2x + 1$.

By replacing all instances of x with $\dfrac{1}{x}$, find another identity that $f(x)$ satisfies.

(c) By solving these two identities simultaneously, express $f(x)$ in terms of x. *[10 marks]*

8 Binomial expansion

In this chapter you will learn:

- how to expand an expression of the form $(a+b)^n$ for positive integer n
- how to find individual terms in the expansion of $(a+b)^n$ for positive integer n
- how to use partial expansions of $(a+bx)^n$ to find approximate values.

Introductory problem

Without using a calculator, find the value of $(1.002)^{10}$ correct to 8 decimal places.

A **binomial** expression is one which contains two terms, for example, $a+b$.

Expanding a power of such a binomial expression could be performed by expanding brackets; for example $(a+b)^7$ could be found by calculating, at length,

$$(a+b)(a+b)(a+b)(a+b)(a+b)(a+b)(a+b).$$

This is time-consuming, but happily there is a much faster approach.

The Ancient Babylonians made an unexpected use of expanding brackets – it helped them multiply numbers in their base 60 number system. This is explored in Supplementary sheet 7 'Babylonian multiplication' on the CD-ROM.

8A Introducing the binomial theorem

To see how to expand an expression of the form $(a+b)^n$ for integer n rapidly, consider the following expansions of $(a+b)^n$, done using the slow method of repeatedly multiplying out brackets.

$$\begin{aligned}
(a+b)^0 &= 1 \\
&= 1a^0b^0 \\
(a+b)^1 &= a+b \\
&= 1a^1b^0 + 1a^0b^1 \\
(a+b)^2 &= a^2+2ab+b^2 \\
&= 1a^2b^0 + 2a^1b^1 + 1a^0b^2 \\
(a+b)^3 &= a^3+3a^2b+3ab^2+b^3 \\
&= 1a^3b^0 + 3a^2b^1 + 3a^1b^2 + 1a^0b^3 \\
(a+b)^4 &= a^4+4a^3b+6a^2b^2+4ab^3+b^4 \\
&= 1a^4b^0 + 4a^3b^1 + 6a^2b^2 + 4a^1b^3 + 1a^0b^4
\end{aligned}$$

The red numbers form a famous mathematical construction called Pascal's triangle. Each value not on the edge is formed by adding up the value directly above and the value to its left. There are many amazing patterns in Pascal's triangle – for example, if you highlight all the even numbers you generate an ever-repeating pattern called a fractal.

See chapter 1, Section D for more combinations $\binom{n}{r}$.

We can see several patterns in this structure.

- The powers of a and b always total n (in the 3rd row, $3+0=2+1=1+2=0+3=3$).
- Each power of a from 0 up to n is present in one of the terms, with the corresponding complementary power of b.
- Each **term** has a **coefficient** (given in red), and the pattern of coefficients in each line is symmetrical.

See Calculator skills sheet 3 on the CD-ROM for a reminder of how to use your calculator to find $\binom{n}{r}$.

In this section we shall focus on the coefficients. The pattern of numbers may seem familiar as they are all numbers which were found when calculating combinations $\binom{n}{r}$. In this context, $\binom{n}{r}$ is called a **binomial coefficient**.

KEY POINT 8.1

Binomial coefficient

The binomial coefficient, $\binom{n}{r}$, is the coefficient of the term containing $a^{n-r}b^r$ in the expansion of $(a+b)^n$.

Worked example 8.1

Find the coefficient of x^5y^3 in the expansion of $(x+y)^8$.

Write down the required term in the form $\binom{n}{r}(a)^{n-r}(b)^r$ with $a=x$, $b=y$, $n=8$, $r=3$

The relevant term is $\binom{8}{3}(x)^5(y)^3$

Calculate the coefficient and apply the powers to the bracketed terms

$\binom{8}{3}=56$

$(x)^5=x^5$

$(y)^3=y^3$

Combine the elements to calculate the coefficient

The term is $56x^5y^3$

The coefficient is 56

EXAM HINT

Questions might ask you to give either the whole term or just the coefficient. Make sure that you answer the question!

Exercise 8A

1. (a) Find the coefficient of xy^3 in the expansion of $(x+y)^4$.
   (b) Find the coefficient of x^3y^4 in the expansion of $(x+y)^7$.
   (c) Find the coefficient of ab^6 in the expansion of $(a+b)^7$.
   (d) Find the coefficient of a^5b^3 in the expansion of $(a+b)^8$.

2. (a) Find the term in x^5y^7 in the expansion of $(x+y)^{12}$.
   (b) Find the term in a^7b^9 in the expansion of $(a+b)^{16}$.
   (c) Find the term in c^3d^2 in the expansion of $(c+d)^5$.
   (d) Find the term in a^2b^7 in the expansion of $(a+b)^9$.
   (e) Find the term in x^2y^4 in the expansion of $(x+y)^6$.

8B Applying the binomial theorem

In Section 8A, you saw the general pattern for expanding powers of a binomial expression $(a+b)$. When expanding powers of more complicated expressions, you will still use this method, but may substitute more complicated expressions for a and b.

Worked example 8.2

Find the term in x^6y^4 in the expansion of $(x+3y^2)^8$.

Write down the required term in the form $\binom{n}{r}(a)^{n-r}(b)^r$ with $a=x$, $b=3y^2$, $n=8$, $r=2$

The relevant term is $\binom{8}{2}(x)^6(3y^2)^2$

Calculate the coefficient and apply the powers to the bracketed terms

$\binom{8}{2}=28$

$(x)^6=x^6$

$(3y^2)^2=9y^4$

The term is $28\times x^6\times 9y^4=252x^6y^4$

Many examination questions ask you to focus on just one term, but you should also be able to find the entire expansion. To do this you repeat the process for each term for every possible value of r (from 0 up to n) and add together the results. This result is quoted in the Formula booklet.

EXAM HINT

Take care to apply the power not only to the algebraic part but also to its coefficient. In Worked example 8.2, $(3y^2)^2 = 9y^4$, not $3y^4$.

KEY POINT 8.2

Binomial theorem

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n$$

Worked example 8.3

Use the Binomial theorem to expand and simplify $(2x-3y)^5$.

Write down each term in the form $\binom{n}{r}(a)^{n-r}(b)^r$ with $a = 2x$, $b = -3y$, $n = 5$

Coefficients: 1, 5, 10, 10, 5, 1

The expansion is

$1(2x)^5 + 5(2x)^4(-3y)^1 + 10(2x)^3(-3y)^2$
$+ 10(2x)^2(-3y)^3 + 5(2x)^1(-3y)^4 + 1(-3y)^5$

Apply the powers to the bracketed terms and multiply through

$= 32x^5 - 240x^4y + 720x^3y^2 - 1080x^2y^3 + 810xy^4 - 243y^5$

EXAM HINT

A common mistake is to assume that the powers of each variable correspond to the value of r in the expansion.

A question may ask for a binomial expansion where both of the terms in the binomial expression contain the same variable. You can use the rules of exponents to determine which term of the expansion is needed.

Worked example 8.4

Find the coefficient of x^5 in the expansion of $\left(2x^2 - \frac{1}{x}\right)^7$.

Start with the form of a general term and simplify using the rules of exponents

Each term will be of the form

$\binom{7}{r}(2x^2)^{7-r}(-x^{-1})^r$

$= \binom{7}{r}(2)^{7-r}x^{14-2r}(-1)^r x^{-r}$

$= \binom{7}{r}(2)^{7-r}(-1)^r x^{14-3r}$

continued . . .

We need the term in x^5, so equate that to the power of x in the general term

Require that $14 - 3r = 5$
$\Leftrightarrow \quad 3r = 9$
$\Leftrightarrow \quad r = 3$

Write down the required term in the form $\binom{n}{r}(a)^{n-r}(b)^r$ with $a = 2x^2$, $b = x^{-1}$, $n = 7$, $r = 3$

The relevant term is $\binom{7}{3}(2x^2)^4(-x^{-1})^3$

Calculate the coefficient and apply the powers to the bracketed terms

$\binom{7}{3} = 35$

$(2x^2)^4 = 16x^8$

$(-x^{-1})^3 = -x^{-3}$

Combine the elements to calculate the coefficient

The term is $35 \times 16x^8 \times (-x^{-3}) = -560x^5$
The coefficient is -560

EXAM HINT

Don't forget that if there is a negative sign it must stay part of the coefficient of the term it is acting upon. Lots of people fall into this trap!

Applying this process in reverse is always quite tricky, since in general you cannot 'undo' the $\binom{n}{r}$ operation. So, you must use the formula for $\binom{n}{r}$, seen in chapter 1, to rewrite it as a polynomial $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

Remind yourself of Key point 1.6 in chapter 1.

Worked example 8.5

The coefficient of x^2 in the expansion of $(1+3x)^n$ is 189. Find n if $n > 0$.

Write down the required term in the form $\binom{n}{r}(a)^{n-r}(b)^r$ with $a = 1$, $b = 3x$, $r = 2$

Required term is
$\binom{n}{2}(1)^{n-2}(3x)^2$

continued . . .

Simplify the coefficient algebraically and apply the powers to the bracketed terms

$$\binom{n}{2} = \frac{n!}{(n-2)!2!} = \frac{n(n-1)}{2}$$

$$(1)^{n-2} = 1$$

$$(3x)^2 = 9x^2$$

Combine these and equate to the given information

$$\frac{9n(n-1)}{2}x^2 \equiv 189x^2$$

Compare coefficients

$$\frac{9n(n-1)}{2} = 189$$

$$9n^2 - 9n = 378$$

$$n^2 - n - 42 = 0$$

$$(n-7)(n+6) = 0$$

$n = 7$ or $n = -6$

but $n > 0$ $\therefore n = 7$

The binomial expansion can be generalised to situations where n is negative or a fraction. This general binomial expansion results in an infinite polynomial.

Exercise 8B

1. (a) Find the coefficient of xy^3 in:
   (i) $(2x+3y)^4$ (ii) $(5x+y)^4$
   (b) Find the term in x^3y^4 in:
   (i) $(x-2y)^7$ (ii) $(y-2x)^7$
   (c) Find the coefficient of a^2b^3 in:
   (i) $\left(2a - \frac{1}{2}b\right)^5$ (ii) $(17a+3b)^5$

2. (a) Find the coefficient of x^2 in the expansion of:
   (i) $\left(x + \frac{1}{x}\right)^8$ (ii) $\left(2x + \frac{1}{\sqrt{x}}\right)^5$
   (b) Find the constant coefficient in the expansion of:
   (i) $(x - 2x^{-2})^9$ (ii) $(x^3 - 2x^{-1})^4$

3. (a) Fully expand and simplify:
   (i) $(2-x)^5$ (ii) $(3+x)^6$

EXAM HINT

The constant coefficient is the term in x^0. It may also be described as the term independent of x.

(b) (i) Find the first three terms in descending powers of x of $(3x+y)^5$.

(ii) Find the first three terms in ascending powers of d of $(2c-d)^4$.

(c) Fully expand and simplify:

(i) $(2x^2-3x)^3$ (ii) $(2x^{-1}+5y)^3$

(d) Fully expand and simplify:

(i) $\left(2z^2+\dfrac{3}{z}\right)^4$ (ii) $\left(3xy+\dfrac{5x}{y}\right)^3$

4. Write in polynomial form:

(a) $\binom{n}{1}$ (b) $\binom{n}{2}$ (c) $\binom{n}{3}$

5. Which term in the expansion of $(x-2y)^5$ has coefficient:

(a) 80 (b) −80 *[6 marks]*

6. Find the coefficient of x^2y^6 in $(3x+2y^2)^5$. *[5 marks]*

7. Find the term in x^5 in $\left(x^2-\dfrac{3}{x}\right)^7$. *[6 marks]*

8. Find the term that is independent of x in the expansion of $\left(2x-\dfrac{5}{x^2}\right)^{12}$. *[6 marks]*

9. The expansion of $(1+3x)^n$ starts with $1+42x\ldots$ Find the value of n. *[4 marks]*

10. The coefficient of x^2 in $(1+2x)^n$ is 264. Find the value of n given that $n\in\mathbb{N}$. *[6 marks]*

11. The coefficient of x^3 in $(1-5x)^n$ is −10 500. Find the value of n given that $n\in\mathbb{N}$. *[5 marks]*

12. The coefficient of x^2 in $(3+2x)^n$ is 20 412. Find the value of n given that $n\in\mathbb{N}$. *[6 marks]*

8C Products of binomial expansions

We may need to work with a product of a binomial and another expression. It is possible to do this by working with the entire expansion.

Worked example 8.6

Use the Binomial theorem to expand and simplify $(5-3x)(2-x)^4$.

Expand $(2-x)^4$. Coefficients are: 1, 4, 6, 4, 1

The expansion is

$$(5-3x)\left[1(2)^4+4(2)^3(-x)+6(2)^2(-x)^2+4(2)^1(-x)^3+1(-x)^4\right]$$

Multiply each term in the second bracket by 5 and then by $-3x$

$$=5\left[16-32x+24x^2-8x^3+x^4\right]$$
$$-3x\left[16-32x+24x^2-8x^3+x^4\right]$$
$$=80-208x+216x^2-112x^3+29x^4-3x^5$$

This method is quite long-winded. In examinations you will usually only be asked to find a small number of terms from such an expression.

Worked example 8.7

Find the coefficient of a^4b^3 in the expansion of $(a-5b)(a+b)^6$.

Split the product into two parts, and treat each separately

$$a(a+b)^6-5b(a+b)^6$$

Decide which term from each expansion is needed to make a^4b^3

For a^4b^3, we need a^3b^3 from the first bracket $(a\times a^3b^3=a^4b^3)$ and a^4b^2 from the second $(b\times a^4b^2=a^4b^3)$

Write down the required terms in the form $\binom{n}{r}(a)^{n-r}(b)^r$

a^4b^3 will arise from

$$a\times\binom{6}{3}(a)^3(b)^3-5b\times\binom{6}{2}(a)^4(b)^2$$

Calculate the coefficient and apply the powers to the bracketed terms

$\binom{6}{3}=20$ $\qquad$ $\binom{6}{2}=15$

$(a)^3=a^3$ $\qquad$ $(a)^4=a^4$

$(b)^3=b^3$ $\qquad$ $(b)^2=b^2$

Combine the elements to calculate the coefficient

The term is $20a^4b^3-75a^4b^3=-55a^4b^3$

The coefficient is -55

As you get more practised, you may not need to split the problem into several parts explicitly; always consider all possible ways that you can multiply to get the required term.

Worked example 8.8

Find the coefficient of x^4 in the expansion of $(1+3x-x^2)(2+x)^5$.

Split the product into three parts, and treat each separately

$1(2+x)^5+3x(2+x)^5-x^2(2+x)^5$

Decide which term from each expansion is needed to make x^4

Write down the required terms in the form $\binom{n}{r}(a)^{n-r}(b)^r$

x^4 will arise from

$1\times\binom{5}{4}(2)^1x^4+3x\times\binom{5}{3}(2)^2x^3-x^2\times\binom{5}{2}(2)^3x^2$

Calculate the coefficient and apply the powers to the bracketed terms

$\binom{5}{4}=5$ $\quad\binom{5}{3}=10$ $\quad\binom{5}{2}=10$

$(2)^1=2$ $\quad(2)^2=4$ $\quad(2)^3=8$

Combine the elements to calculate the coefficient

The term is

$1\times5\times2x^4+3x\times10\times4x^3-x^2\times10\times8x^2$

$=10x^4+120x^4-80x^4$

$=50x^4$

The coefficient is 50

Exercise 8C

1. (a) (i) Find the coefficient of x^2y^5 in the expansion of $(x-y)(x+y)^6$.

(ii) Find the coefficient of x^5 in the expansion of $(1+3x)(1+x)^7$.

(b) (i) Find the coefficient of x^6 in the expansion of $(1-x^2)(1+x)^5$.

(ii) Find the coefficient of x^6 in the expansion of $(1-x^2)(1+x)^7$.

2. (a) (i) Find the coefficient of x^2y in the expansion of $(1+x)^3(1+y)^5$.